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(54) **SYSTEM AND METHOD FOR COMPUTER BASED RISK CONTROL ANALYSIS**

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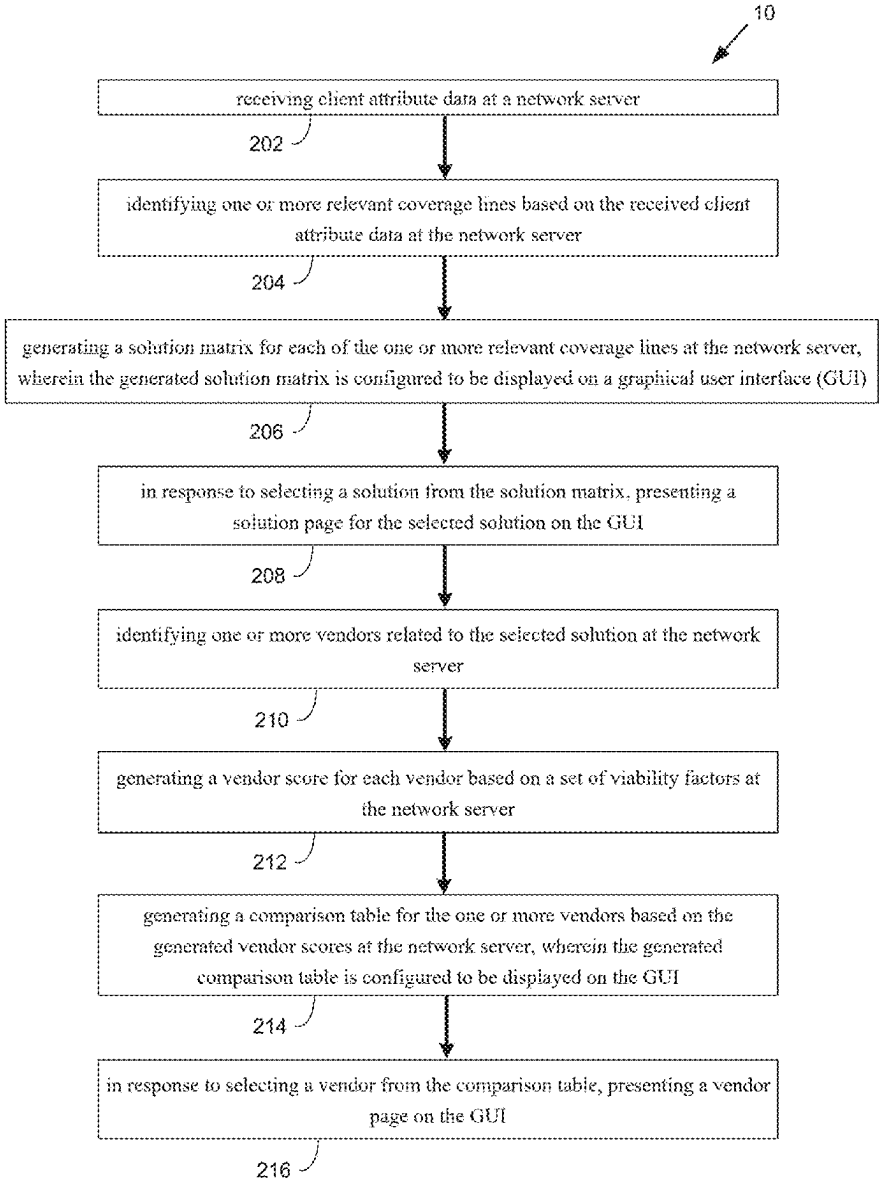
G06F 3/0483 (2013.01)

(52) **U.S. Cl.**
CPC ***G06Q 40/08*** (2013.01); ***G06F 3/0483*** (2013.01)

(57) **ABSTRACT**

A method, computer program product, and computing system for receiving client attribute data, identifying one or more relevant coverage lines based on the received client attribute data, and generating a solution matrix for each of the one or more relevant coverage lines.



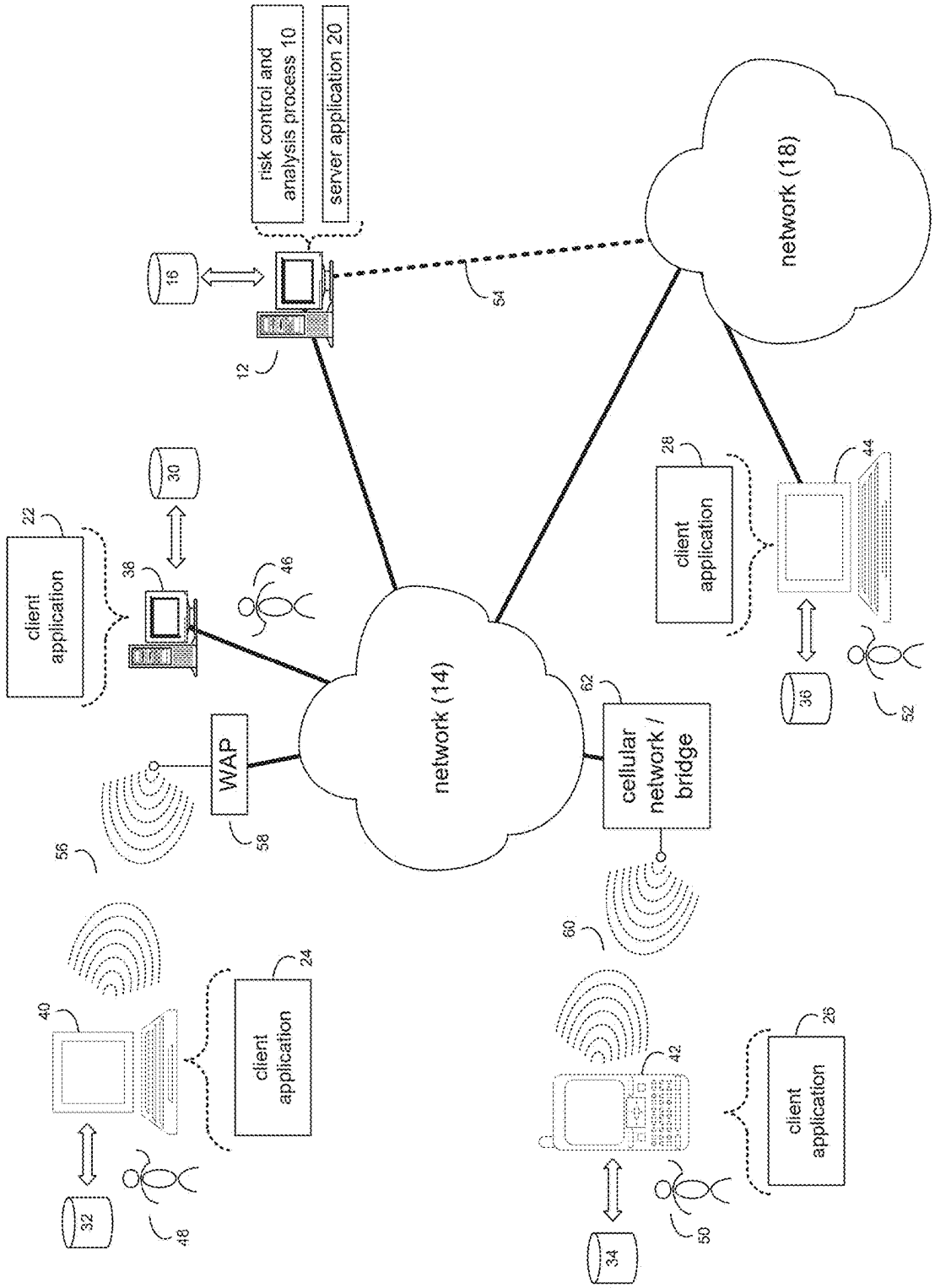

risk control and analysis process 10
server application 20
16
12
54
network (18)
30
22
client application
38
46
network (14)
28
client application
44
36
52
62
cellular network / bridge
WAP
58
56
60
40
24
client application
32
48
42
26
client application
34
50

FIG. 1

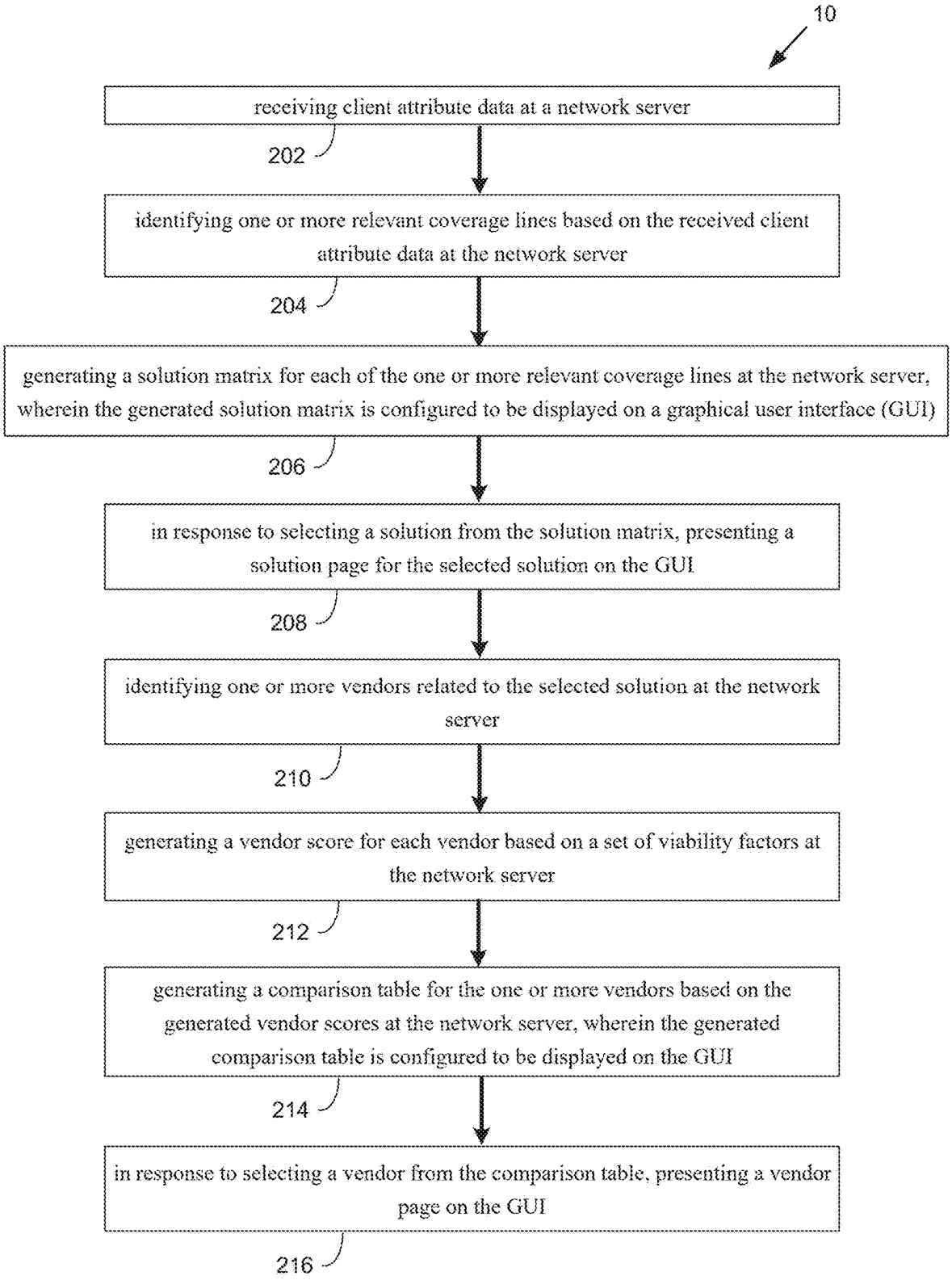

10
receiving client attribute data at a network server
202
identifying one or more relevant coverage lines based on the received client attribute data at the network server
204
generating a solution matrix for each of the one or more relevant coverage lines at the network server, wherein the generated solution matrix is configured to be displayed on a graphical user interface (GUI)
206
in response to selecting a solution from the solution matrix, presenting a solution page for the selected solution on the GUI
208
identifying one or more vendors related to the selected solution at the network server
210
generating a vendor score for each vendor based on a set of viability factors at the network server
212
generating a comparison table for the one or more vendors based on the generated vendor scores at the network server, wherein the generated comparison table is configured to be displayed on the GUI
214
in response to selecting a vendor from the comparison table, presenting a vendor page on the GUI
216

FIG. 2

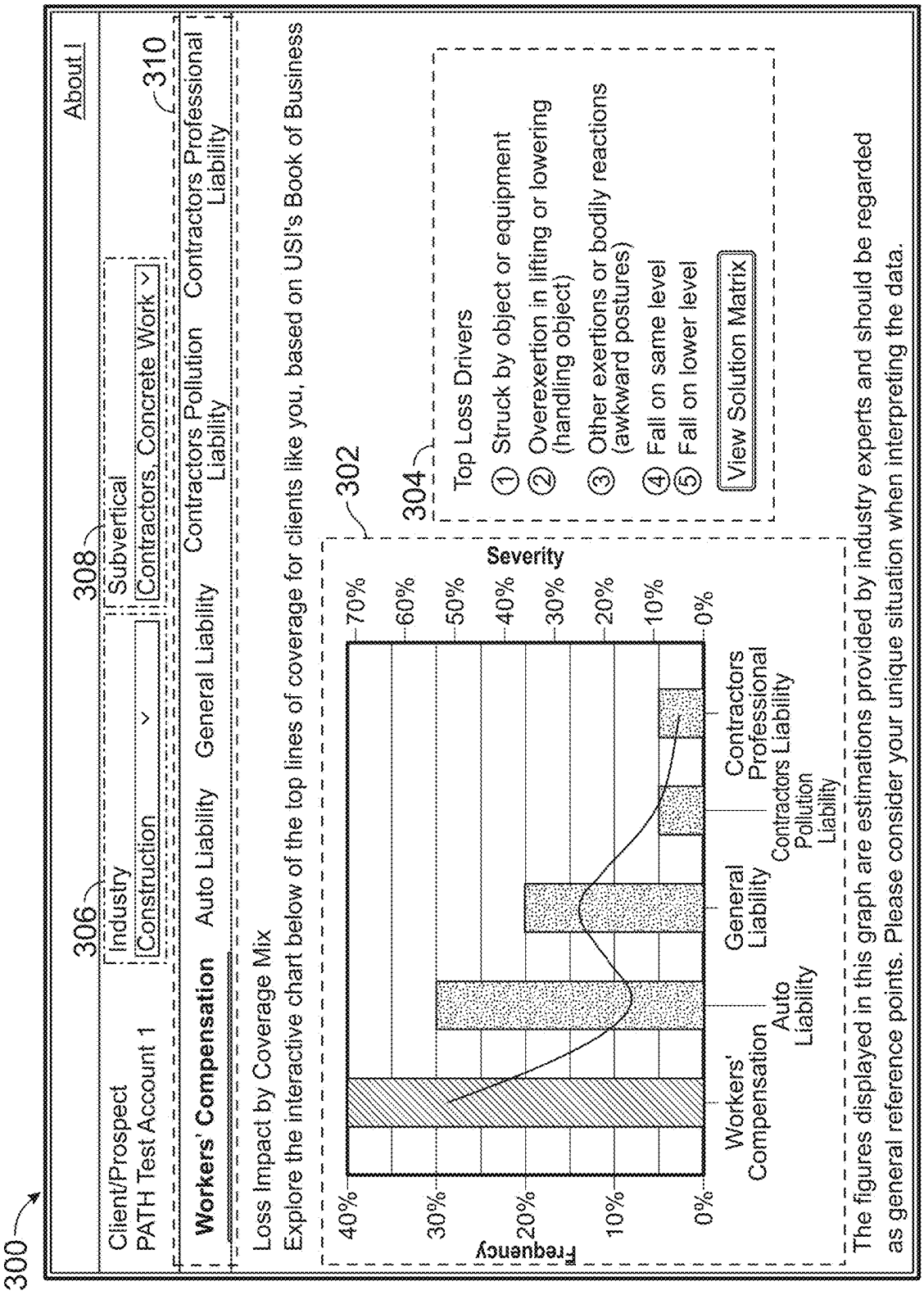

300
About I
Client/Prospect
PATH Test Account 1
306
Industry
Construction
308
Subvertical
Contractors, Concrete Work
310
Workers' Compensation
Auto Liability
General Liability
Contractors Pollution Liability
Contractors Professional Liability
Loss Impact by Coverage Mix
Explore the interactive chart below of the top lines of coverage for clients like you, based on USI's Book of Business
302
Frequency
40%
30%
20%
10%
0%
Severity
70%
60%
50%
40%
30%
20%
10%
0%
Workers' Compensation
Auto Liability
General Liability
Contractors Pollution Liability
Contractors Professional Liability
304
Top Loss Drivers
① Struck by object or equipment
② Overexertion in lifting or lowering (handling object)
③ Other exertions or bodily reactions (awkward postures)
④ Fall on same level
⑤ Fall on lower level
View Solution Matrix
The figures displayed in this graph are estimations provided by industry experts and should be regarded as general reference points. Please consider your unique situation when interpreting the data.

FIG. 3

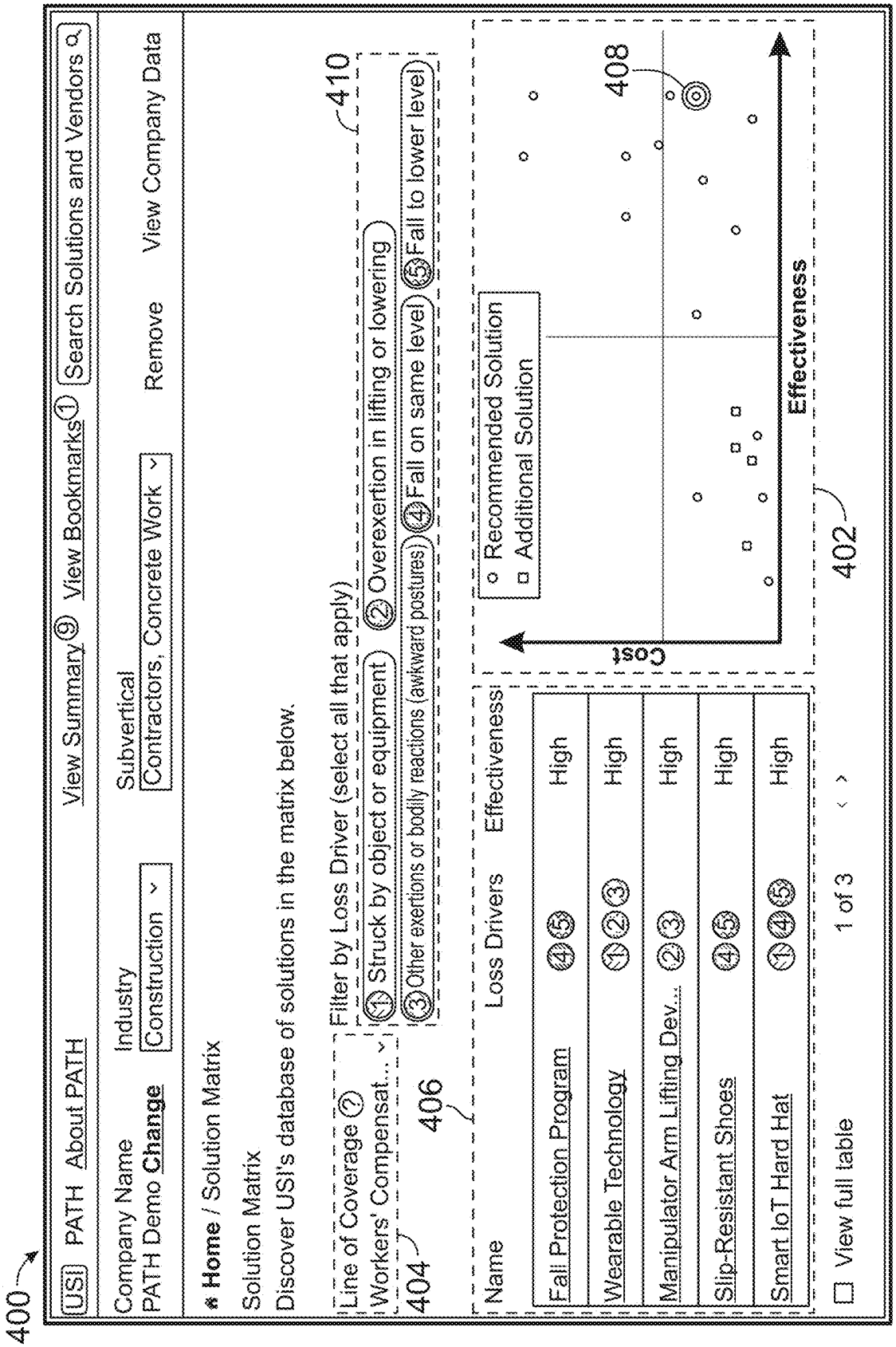

400
USI PATH About PATH
View Summary 9 View Bookmarks 1 Search Solutions and Vendors
Company Name
PATH Demo Change
Industry
Construction
Subvertical
Contractors, Concrete Work
Remove
View Company Data
Home / Solution Matrix
Solution Matrix
Discover USI's database of solutions in the matrix below.
Line of Coverage
Workers' Compensat...
404
406
Filter by Loss Driver (select all that apply)
1 Struck by object or equipment
2 Overexertion in lifting or lowering
3 Other exertions or bodily reactions (awkward postures)
4 Fall on same level
5 Fall to lower level
410
Name
Loss Drivers
Effectiveness
Fall Protection Program 4 5 High
Wearable Technology 1 2 3 High
Manipulator Arm Lifting Dev... 2 3 High
Slip-Resistant Shoes 4 5 High
Smart IoT Hard Hat 1 4 5 High
View full table
1 of 3
Recommended Solution
Additional Solution
Cost
Effectiveness
408
402

FIG. 4

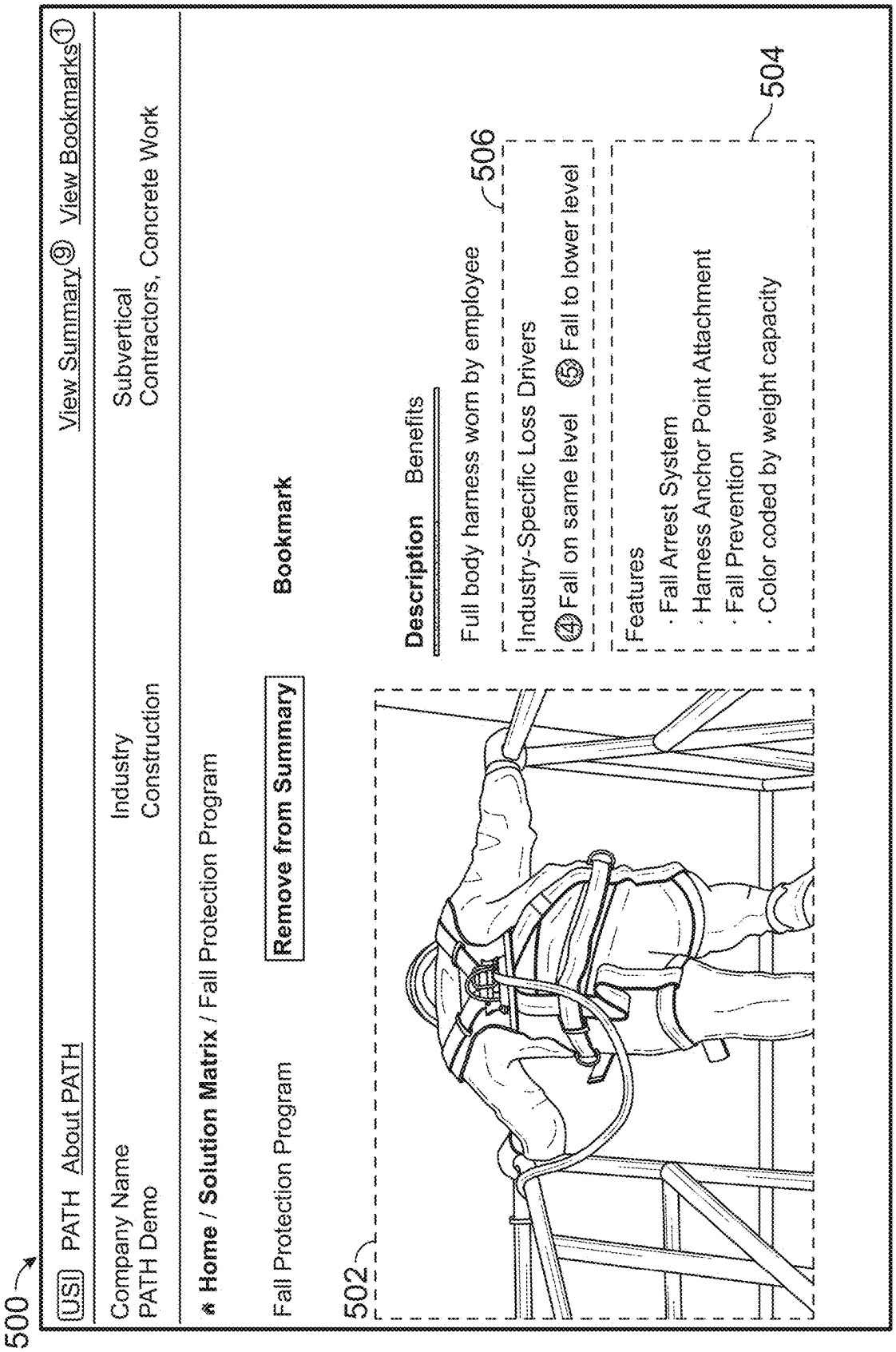

500
USI PATH About PATH
View Summary 9 View Bookmarks 1
Company Name
PATH Demo
Industry
Construction
Subvertical
Contractors, Concrete Work
Home / Solution Matrix / Fall Protection Program
Fall Protection Program
Remove from Summary
Bookmark
502
Description Benefits
Full body harness worn by employee
506
Industry-Specific Loss Drivers
4 Fall on same level
5 Fall to lower level
Features
· Fall Arrest System
· Harness Anchor Point Attachment
· Fall Prevention
· Color coded by weight capacity
504

FIG. 5

600

Calculator: Year 1 ROI

Based on your company's loss history and solution needs, adjust the inputs below to understand this solution's Year 1 ROI.

RETURN		INVESTMENT	
Estimated Losses ?	2	Cost Per Unit ?	$ 200
÷ Every ?	1 year(s)	× # of Units ?	60
× Industry Avg Cost Per Loss ?	$ 51,000	× USI Discount ☐	0%
= **Total Annual Losses**	**$ 102,000**	= **Total Implementation Costs**	**$ 12,000**
× Solution Impact ?	80%	Ongoing Costs ?	$ 1,200
+ Indirect Cost Savings ? ☐	$		
= **Year 1 Savings**	**$ 81,600**		

6.8x Year 1 ROI ?

For every $1 you invest, you have the potential to save $6.80 in claim cost reductions

☐ View as Percentage

Disclaimer: Savings estimates are illustrative given data limitations and are subject to change based on your particular circumstances and market conditions.

FIG. 6

700

Projection: Your 5-Year ROI
Customize the inputs below to see this solution's 5-year impact.

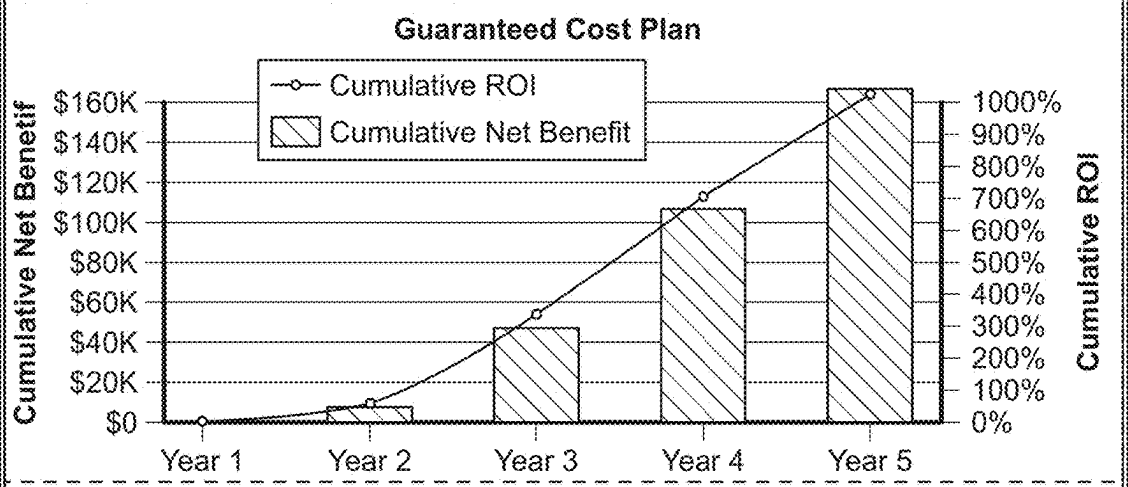
Guaranteed Cost Deductible

Carrier Credits ?

0%

702

☐ Hide graph



Detailed View Summary View

Category		Y1	Y2	Y3	Y4	Y5
Investment	Implementation Costs ?	$12,000	-	-	-	-
	Ongoing Costs ?	-	$1,200	$1,200	$1,200	$1,200
	Cumulative Costs ?	$12,000	$13,200	$14,400	$15,600	$16,800
Return	Premium Savings ?	-	$20,400	$40,800	$61,200	$61,200
	Out Of Pocket Claims Avoidance ?	-	-	-	-	-
	Cumulative Savings ?	-	$20,400	$61,200	$122,400	$183,600
	Cumulative Net Benefit ?	-$12,000	$7,200	$46,800	$106,800	$166,800
	Cumulative ROI ?	-	55%	325%	685%	993%
	Internal Rate of Return (IRR) ?					151%

FIG. 7

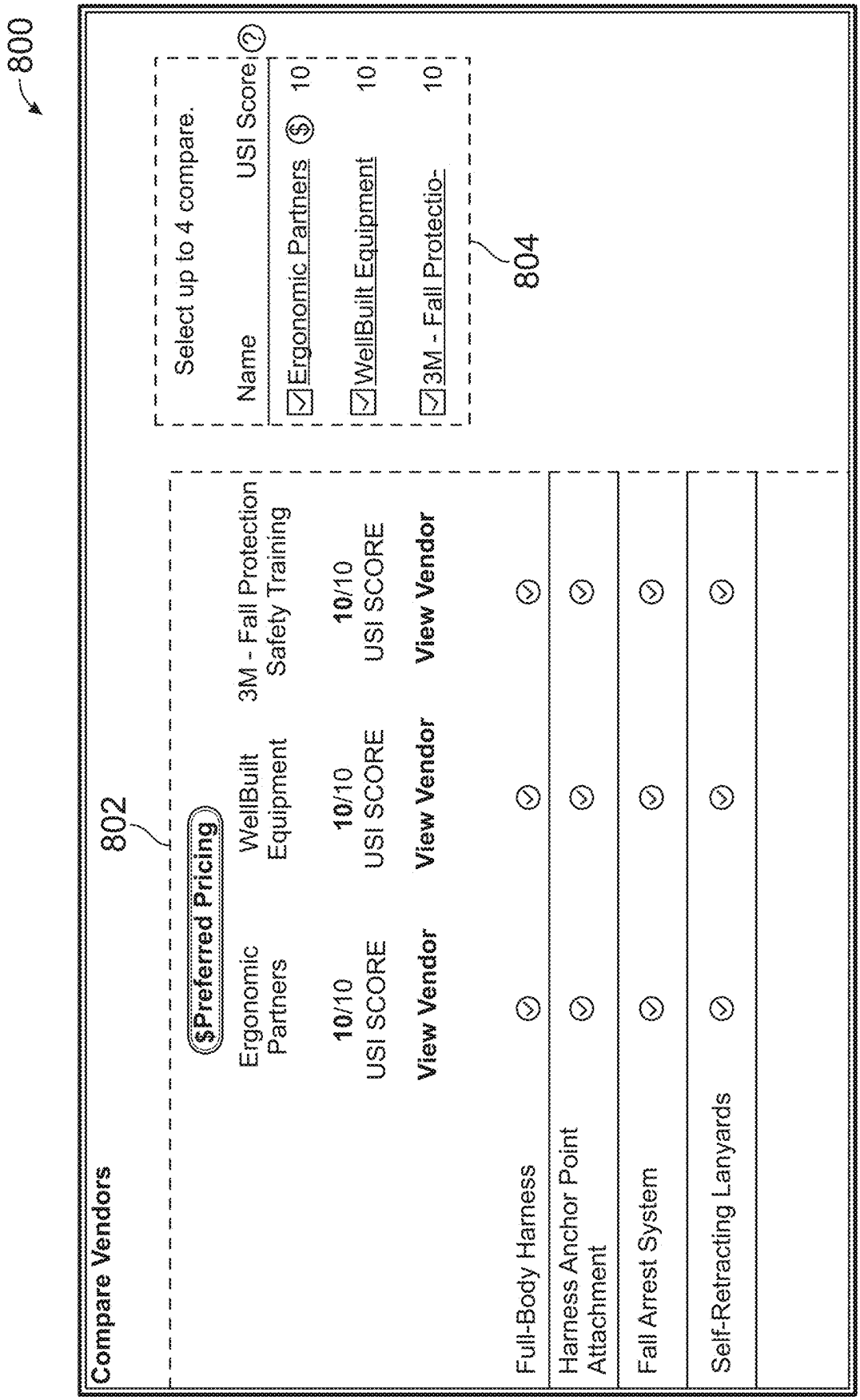

800
Compare Vendors
802
$Preferred Pricing
Ergonomic Partners
10/10
USI SCORE
View Vendor
WellBuilt Equipment
10/10
USI SCORE
View Vendor
3M - Fall Protection Safety Training
10/10
USI SCORE
View Vendor
Full-Body Harness
Harness Anchor Point Attachment
Fall Arrest System
Self-Retracting Lanyards
Select up to 4 compare.
Name
USI Score
Ergonomic Partners $ 10
WellBuilt Equipment 10
3M - Fall Protectio- 10
804

FIG. 8

900

Home / **Solution Matrix** / **Fall Protection Program** / Ergonomic Partners

Ergonomic Partners **Remove from Summary** **Bookmark**

Ergonomic Partners is an ergonomic material handling solutions company specializing in improving worker productivity while reducing the risk and associated costs of worker injury. With over 50 years of material handling experience, the company combines the best off-the-shelf ergonomic material handling and workstation equipment. Its custom designed and engineered lifting devices and special equipment solutions help to create optimum cost effective improvements for demanding projects.

https://www.ergonomicpartners.com

902

Industry	Subvertical	Line of Coverage
Construction	Contractors, Concrete Work	Workers' Compensation

3 Solutions

Workers' Compensation **Fall Protection Program** **10**/10 **USI Score** **+ $200** per EE system (starting) View features & notes	Workers' Compensation **Manipulator Arm Lifting D...** **10**/10 USI SCORE + $25,000 starting cost per device View features & notes	Workers' Compensation **Heavy Equipment Safety Tr...** **10**/10 USI SCORE + $200 per EE system (starting) View features & Notes

904

Access USI Preferred Pricing

1. Visit our website at **www.ergonomicpartners.com** to view our distributor solutions.
2. Send an email to **ncameron@ergonomicpartners.com** and indicate USI client to receive a 5% discount on distributor products. Discount does not apply to custom-engineered solutions.
3. Nik will reach out to discuss the details of your project and finalize pricing, delivery and Installation.

906

Contacts

Nik Cameron

Sales Development Partner

ncameron@ergonomicpartners.com

908

FIG. 9

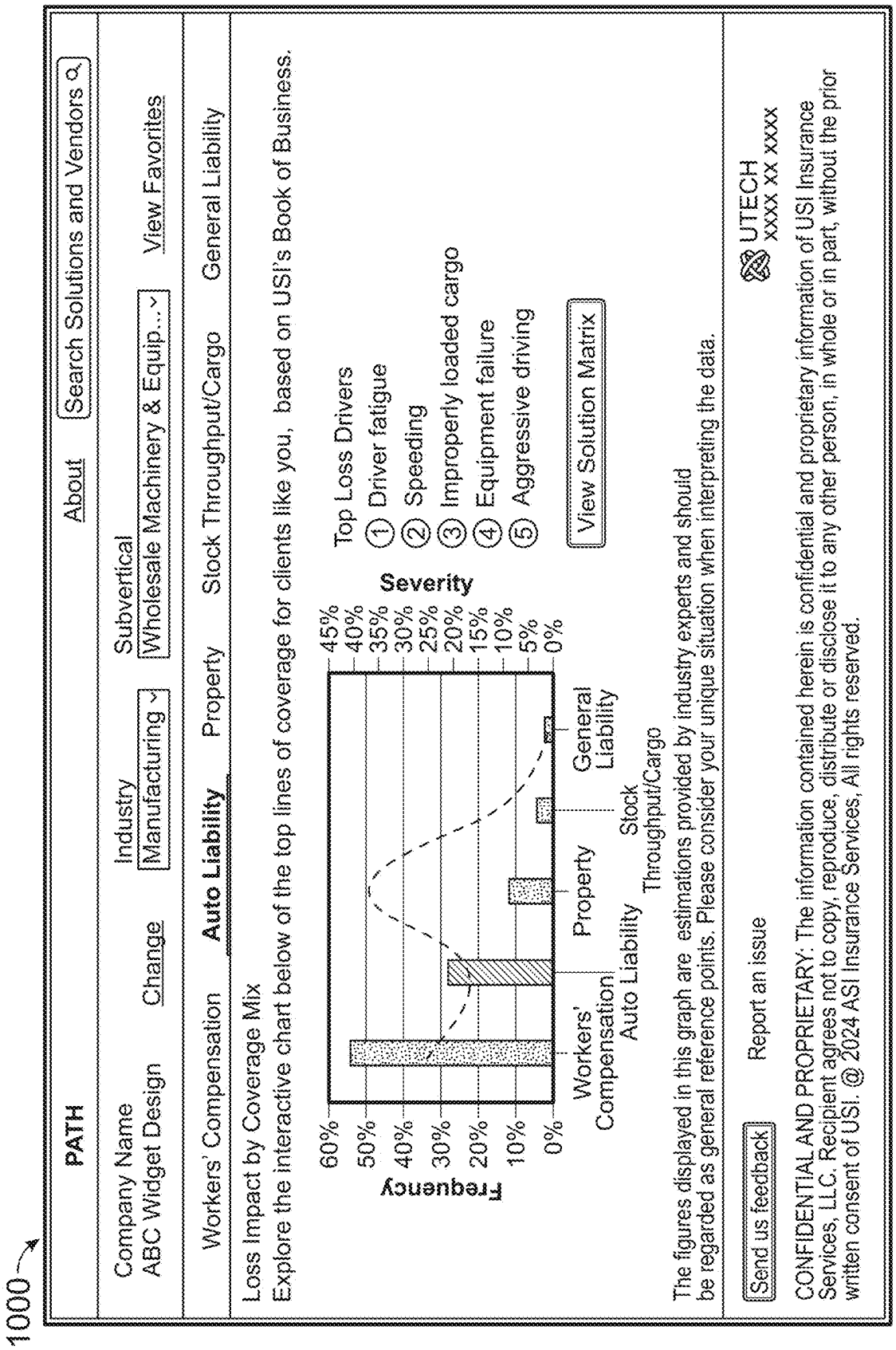

1000
PATH
About
Search Solutions and Vendors
Company Name
ABC Widget Design
Change
Industry
Manufacturing
Subvertical
Wholesale Machinery & Equip...
View Favorites
Workers' Compensation
Auto Liability
Property
Stock Throughput/Cargo
General Liability
Loss Impact by Coverage Mix
Explore the interactive chart below of the top lines of coverage for clients like you, based on USI's Book of Business.
Frequency
60%
50%
40%
30%
20%
10%
0%
Severity
45%
40%
35%
30%
25%
20%
15%
10%
5%
0%
Workers' Compensation
Auto Liability
Property
Stock Throughput/Cargo
General Liability
Top Loss Drivers
1 Driver fatigue
2 Speeding
3 Improperly loaded cargo
4 Equipment failure
5 Aggressive driving
View Solution Matrix
The figures displayed in this graph are estimations provided by industry experts and should be regarded as general reference points. Please consider your unique situation when interpreting the data.
Send us feedback
Report an issue
UTECH
XXXX XX XXXX
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FIG. 10

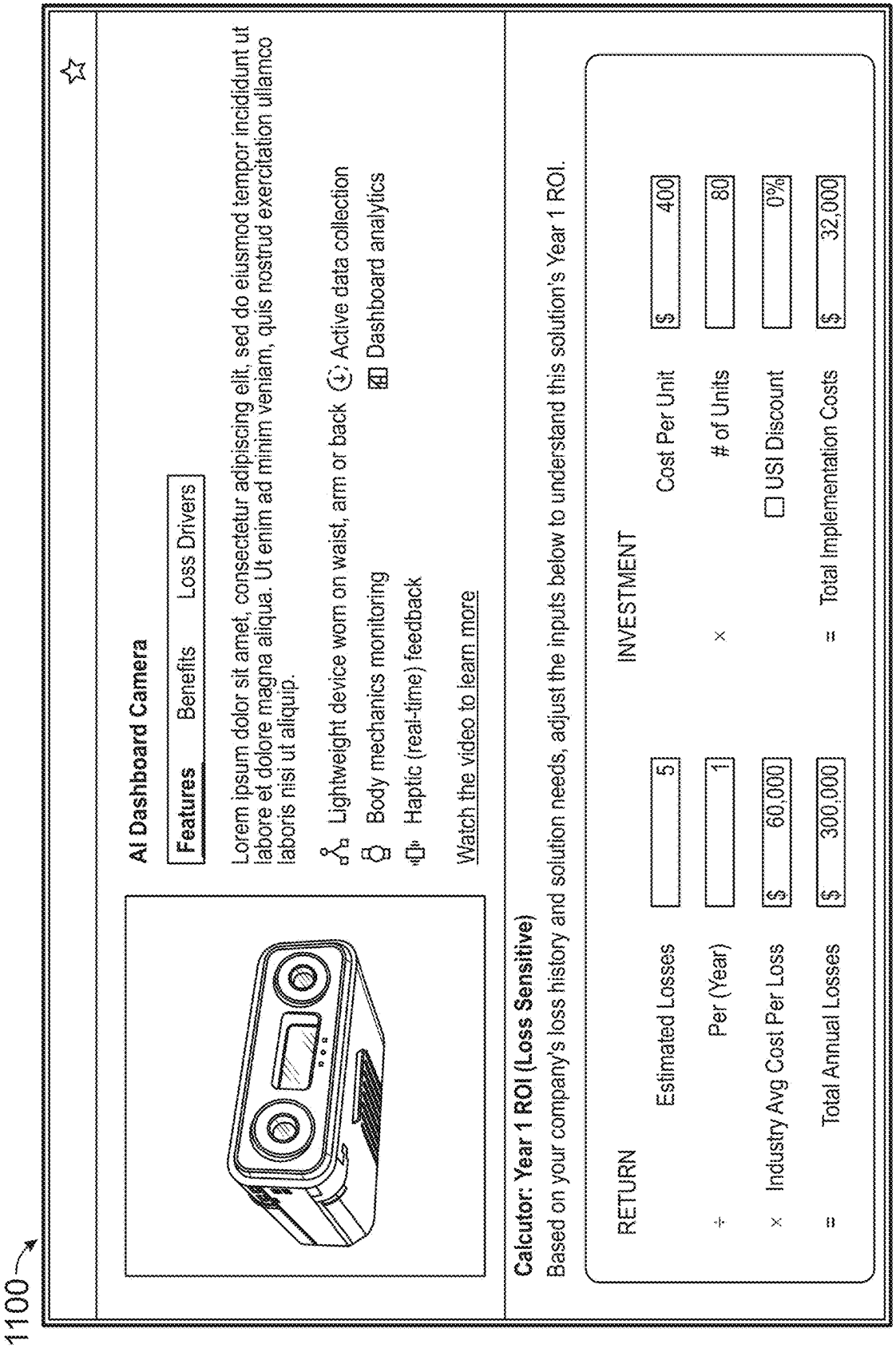

1100
AI Dashboard Camera
Features
Benefits
Loss Drivers
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip.
Lightweight device worn on waist, arm or back
Active data collection
Body mechanics monitoring
Dashboard analytics
Haptic (real-time) feedback
Watch the video to learn more
Calcutor: Year 1 ROI (Loss Sensitive)
Based on your company's loss history and solution needs, adjust the inputs below to understand this solution's Year 1 ROI.
RETURN
Estimated Losses
5
÷
Per (Year)
1
×
Industry Avg Cost Per Loss
$ 60,000
=
Total Annual Losses
$ 300,000
INVESTMENT
Cost Per Unit
$ 400
×
# of Units
80
USI Discount
0%
=
Total Implementation Costs
$ 32,000

FIG. 11

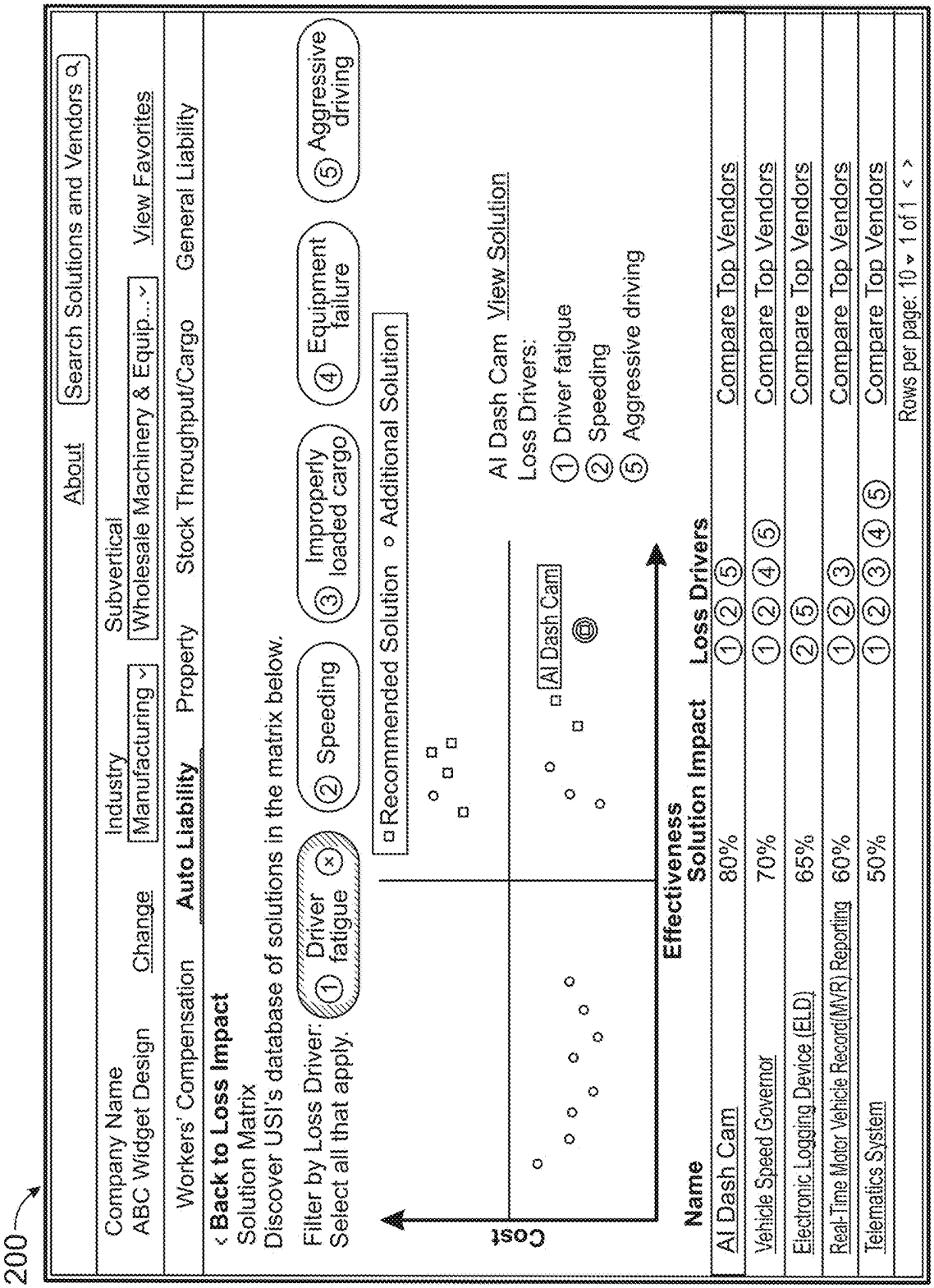
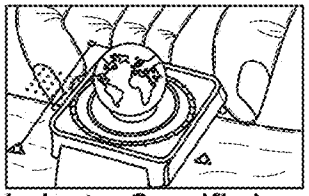

1200
About
Search Solutions and Vendors
Company Name
ABC Widget Design
Change
Industry
Manufacturing
Subvertical
Wholesale Machinery & Equip...
View Favorites
Workers' Compensation
Auto Liability
Property
Stock Throughput/Cargo
General Liability
< Back to Loss Impact
Solution Matrix
Discover USI's database of solutions in the matrix below.
Filter by Loss Driver:
Select all that apply.
① Driver fatigue
② Speeding
③ Improperly loaded cargo
④ Equipment failure
⑤ Aggressive driving
Recommended Solution
Additional Solution
Cost
Effectiveness
AI Dash Cam
AI Dash Cam View Solution
Loss Drivers:
① Driver fatigue
② Speeding
⑤ Aggressive driving
Name
Solution Impact
Loss Drivers
AI Dash Cam 80% ① ② ⑤ Compare Top Vendors
Vehicle Speed Governor 70% ① ② ④ ⑤ Compare Top Vendors
Electronic Logging Device (ELD) 65% ② ⑤ Compare Top Vendors
Real-Time Motor Vehicle Record(MVR) Reporting 60% ① ② ③ Compare Top Vendors
Telematics System 50% ① ② ③ ④ ⑤ Compare Top Vendors
Rows per page: 10 1 of 1 < >

FIG. 12

1300

Wearable Technology ☆

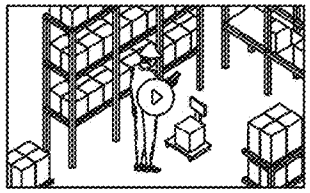


Industry Specific Loss Drivers

- Struck by or against machinery/equipment
- Struck - Moving or handling machinery/equipment
- Strains - Assembly Line Repetitive Motion
- Caught in or compressed by machinery/equipment

Features Benefits

- Lightweight device worn on waist, arm or back
- Body mechanics monitoring
- Haptic (real-time) feedback
- Active data collection
- Dashboard analytics



Calculator: Year 1 ROI (Loss Sensitive)

Based on your company's loss history and solution needs, adjust the inputs below to understand this solution's Year 1 ROI,

RETURN

	Estimated Losses	5
÷	Per (Year)	1
×	Industry Avg Cost Per Loss	$ 60,000
=	Total Annual Losses	$ 300,000
×	Solution Impact	60%
+	☐ Indirect Cost Savings	$ 0
=	Year 1 Savings (Loss Sensitive)	$ 180,000

Reset

INVESTMENT

	Cost Per Unit	$ 400
×	# of Units	80
	☐ USI Discount	0%

= Total Implementation Costs $ 32,000

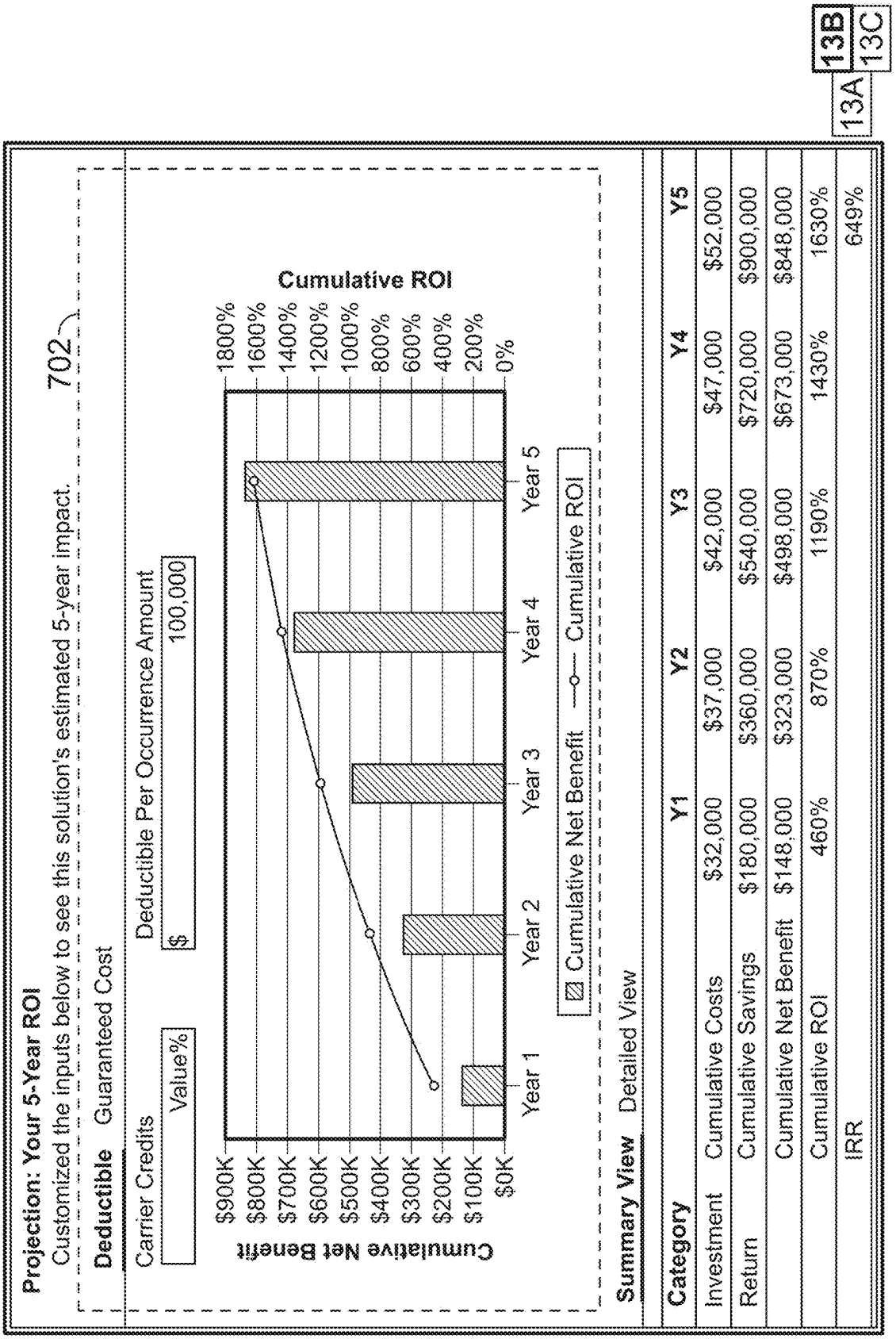
5.6x Year 1 ROI (Loss Sensitive)
For every $1 you invest, you have thepotential to save $5.60 in claim cost reductions

☐ View as percentage

Disclaimer: Savings estimates are illustrative given data limitations and are subject to change based on your particular circumstances and marketconditions.

13A 13B 13C

FIG. 13A



Category		Y1	Y2	Y3	Y4	Y5
Investment	Cumulative Costs	$32,000	$37,000	$42,000	$47,000	$52,000
Return	Cumulative Savings	$180,000	$360,000	$540,000	$720,000	$900,000
	Cumulative Net Benefit	$148,000	$323,000	$498,000	$673,000	$848,000
	Cumulative ROI	460%	870%	1190%	1430%	1630%
	IRR					649%

FIG. 13B

Vendors

	Preferred Pricing Modjoul USI Score 8.5/10 View Vendor	Preferred Pricing Blackline Interna... USI Score 5.3/10 View Vendor	Preferred Pricing Triumvirate USI Score 5.3/10 View Vendor	StrongArm USI Score 4.9/10 View Vendor
Active data collection	✓	✓	✓	✓
Dashboard analytics	✓	✓	✓	✓
Real-time haptic feedback	✓	✗	✓	✓
Body mechanics monitoring	✓	✓	✗	✓
Fall prevention	✗	✓	✗	✗
Fall detection	✓	✓	✓	✓
SOS alerts	✓	✗	✓	✗
Air monitoring	✓	✗	✗	✗
Noise monitoring	✓	✓	✗	✗
Proximity detection/alarm	✓	✓	✓	✓

Compare Vendors
Select up to 4.

	Name	USI Score ⓘ
☑	Modjoul	8.5
☑	Blackline Safety	5.3
☑	Reactec	5.3
☑	StrongArm	4.9
☐	Makusafe	4.5
☐	triax	4.0
☐	Kwant	3.5
☐	Kinetic	3.4
☐	SoterAnalytics	3.4
☐	Kenzen	3.2
☐	Proxxi	3.1
☐	German Bionic	3.0
☐	dorsaVi	2.7
☐	Longitude6	2.6

13A 13B 13C

FIG. 13C

1400

Monnit ☆

Monnit

Monnit is a leading sensor solutions company dedicated to providing innovative, reliable, and customizable monitoring solutions. Leveraging cutting-edge technology. Monnit specializes in designing and manufacturing a diverse range of sensors for various industries. These sensors enable real-time data collection and monitoring, offering businesses and individuals the ability to enhance operational efficiency, ensure safety, and make informed decisions.

www.monnit.com

Solutions

Industry	Subvertical	Line of Coverage
Manufacturing	Wholesale Machinery & Equipment	All Lines of Coverage

4 Solutions

Property	Property	Stock throughput/Cargo	Property
Condition Monitoring Solution	**IoT Roof Drainage Sensor**	**Aspiring Smoke Detector (ASD)**	**Flood/Water Detection Sensor**
9.7/10	8.7/10	8.0/10	7.8/10
$140 per device	$100 per sensor	$75 per device	$90 per sensor
• $95 per location per year	• $95 per location per year	• $10 estimated maintenance per year	• $45 per location per year
• $650 per gateway	• $650 per gateway		• $650 per gateway
View features	View features	View features	View features

Next Steps

1. Visit Monnit's special USI page (monnit.com/usi-special-offer) to unlock a 10% discount offered to USI clients.
2. Complete the demo request form.
3. Await confirmation from Monnit's customer support team to schedule a demo.
4. Review your requirements during your demo with Monnit, where they will share more about their solutions and pricing.

Contacts

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123-456-7890

Sarah Essich
Strategic Partnership Rep
sarah@modjoul.com
123-456-7890

FIG. 14

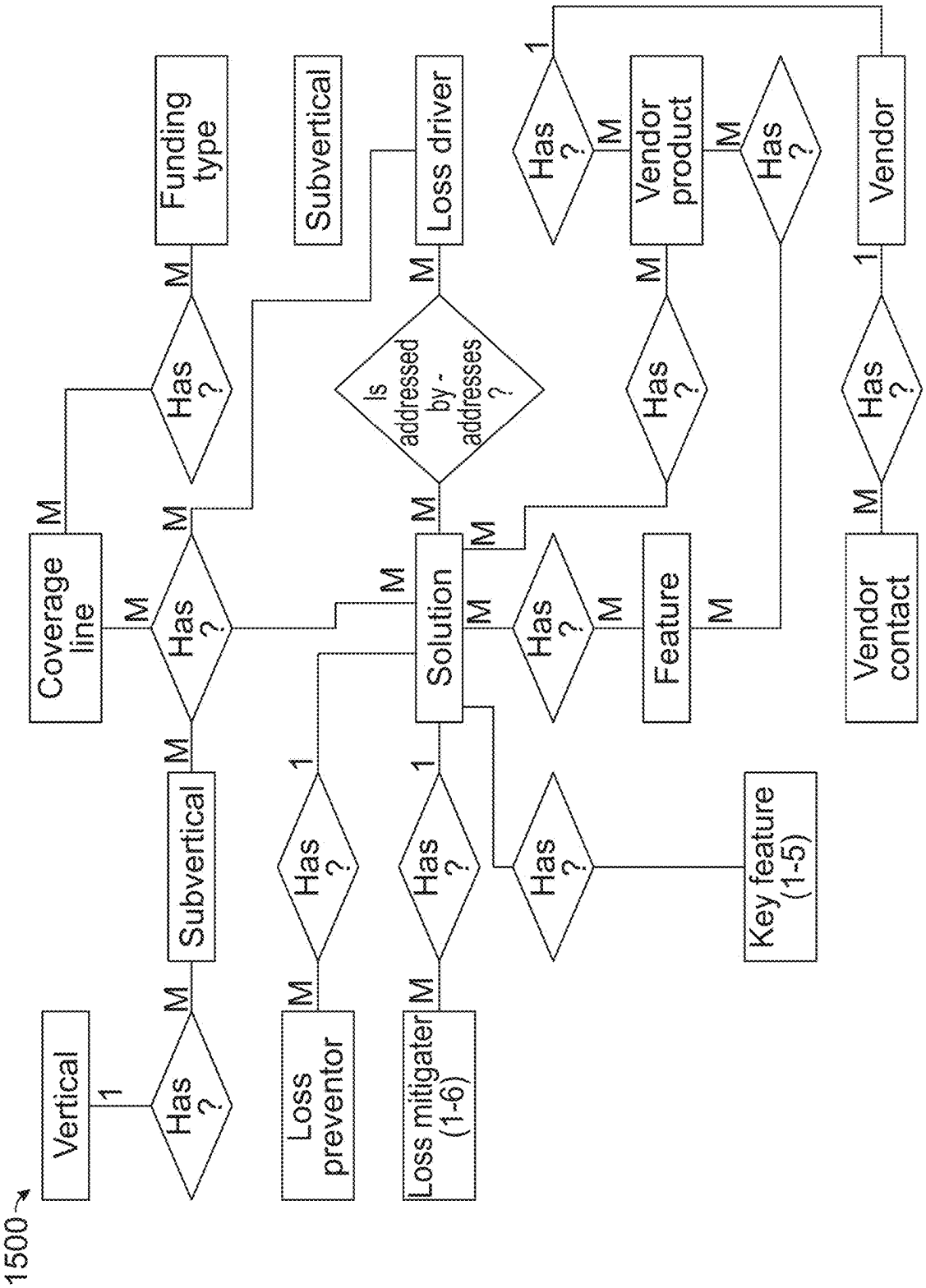

1500
Vertical
Has ?
Subvertical
Coverage line
Has ?
Funding type
Has ?
Subvertical
Loss preventor
Has ?
Loss mitigater (1-6)
Has ?
Solution
Loss driver
Is addressed by - addresses ?
Has ?
Key feature (1-5)
Has ?
Feature
Has ?
Vendor product
Has ?
Has ?
Vendor contact
Has ?
Vendor
1
M

FIG. 15

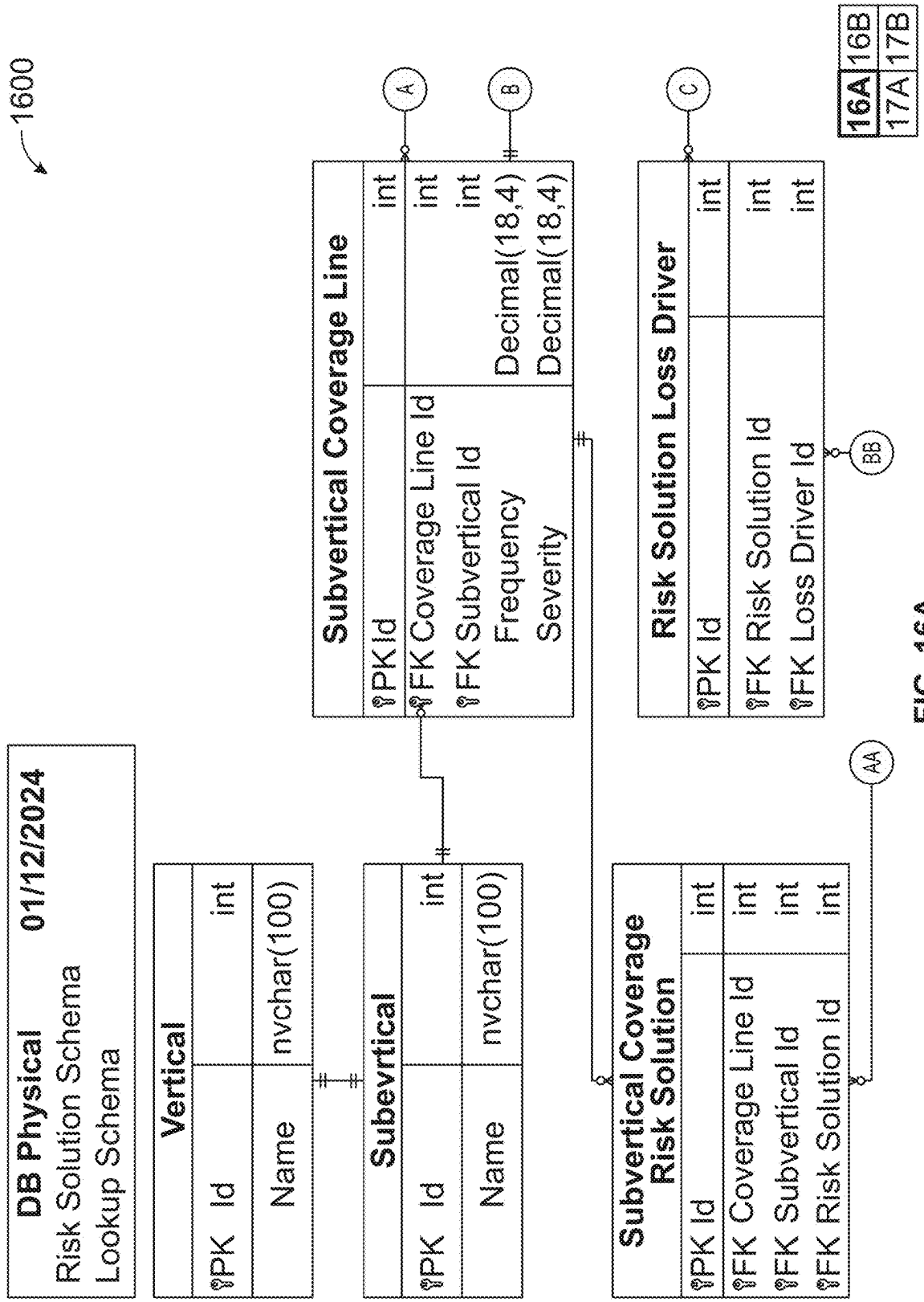

1600
DB Physical 01/12/2024
Risk Solution Schema
Lookup Schema
Vertical
PK Id int
Name nvchar(100)
Subevrtical
PK Id int
Name nvchar(100)
Subvertical Coverage Line
PK Id int
FK Coverage Line Id int
FK Subvertical Id int
Frequency Decimal(18,4)
Severity Decimal(18,4)
Subvertical Coverage Risk Solution
PK Id int
FK Coverage Line Id int
FK Subvertical Id int
FK Risk Solution Id int
Risk Solution Loss Driver
PK Id int
FK Risk Solution Id int
FK Loss Driver Id int
A
B
C
AA
BB
16A 16B
17A 17B

FIG. 16A

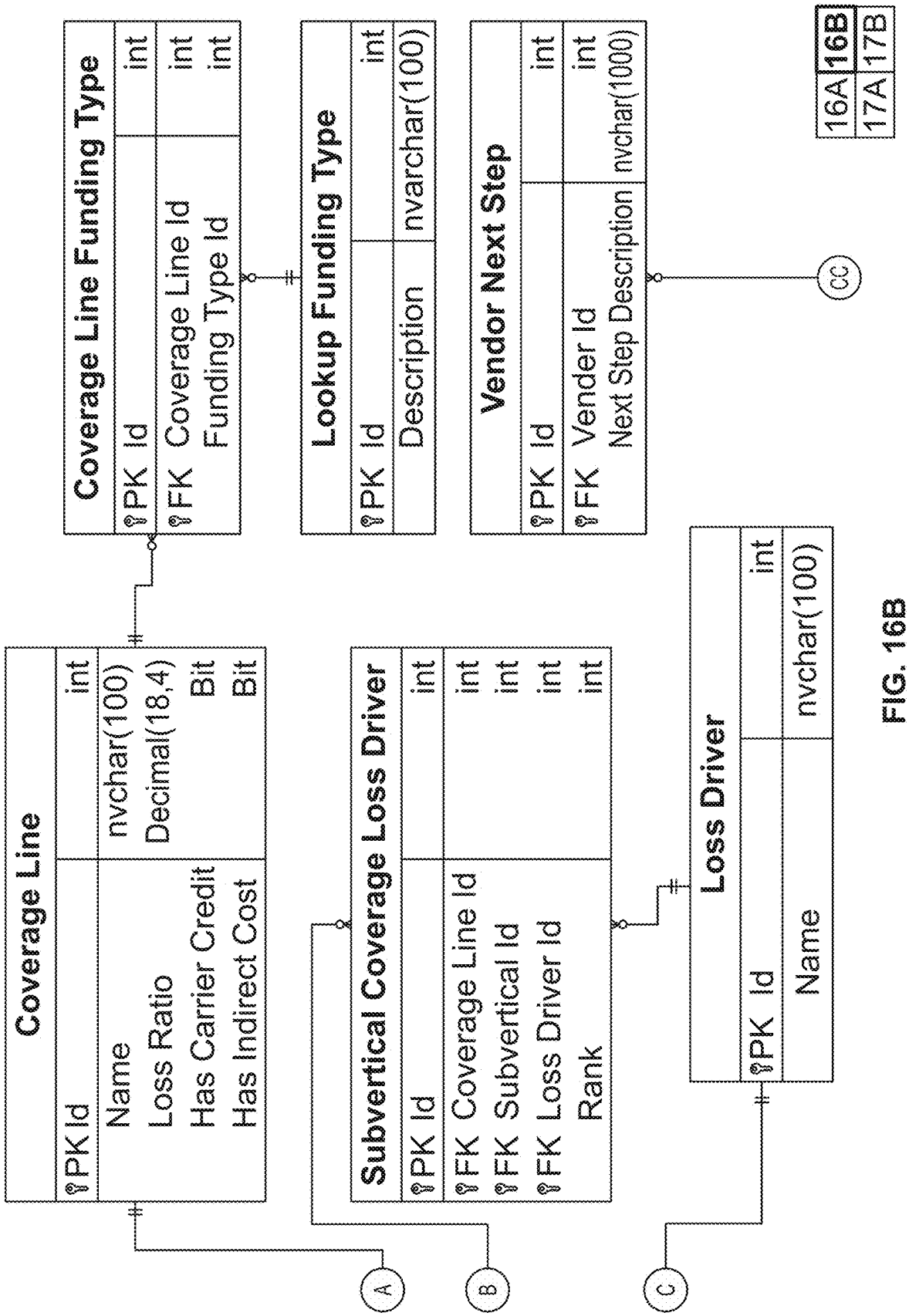

Coverage Line Funding Type
PK Id int
FK Coverage Line Id int
Funding Type Id int
Lookup Funding Type
PK Id int
Description nvarchar(100)
Vendor Next Step
PK Id int
FK Vender Id int
Next Step Description nvchar(1000)
16A 16B
17A 17B
CC
Coverage Line
PK Id int
Name nvchar(100)
Loss Ratio Decimal(18,4)
Has Carrier Credit Bit
Has Indirect Cost Bit
Subvertical Coverage Loss Driver
PK Id int
FK Coverage Line Id int
FK Subvertical Id int
FK Loss Driver Id int
Rank int
Loss Driver
PK Id int
Name nvchar(100)
A
B
C

FIG. 16B

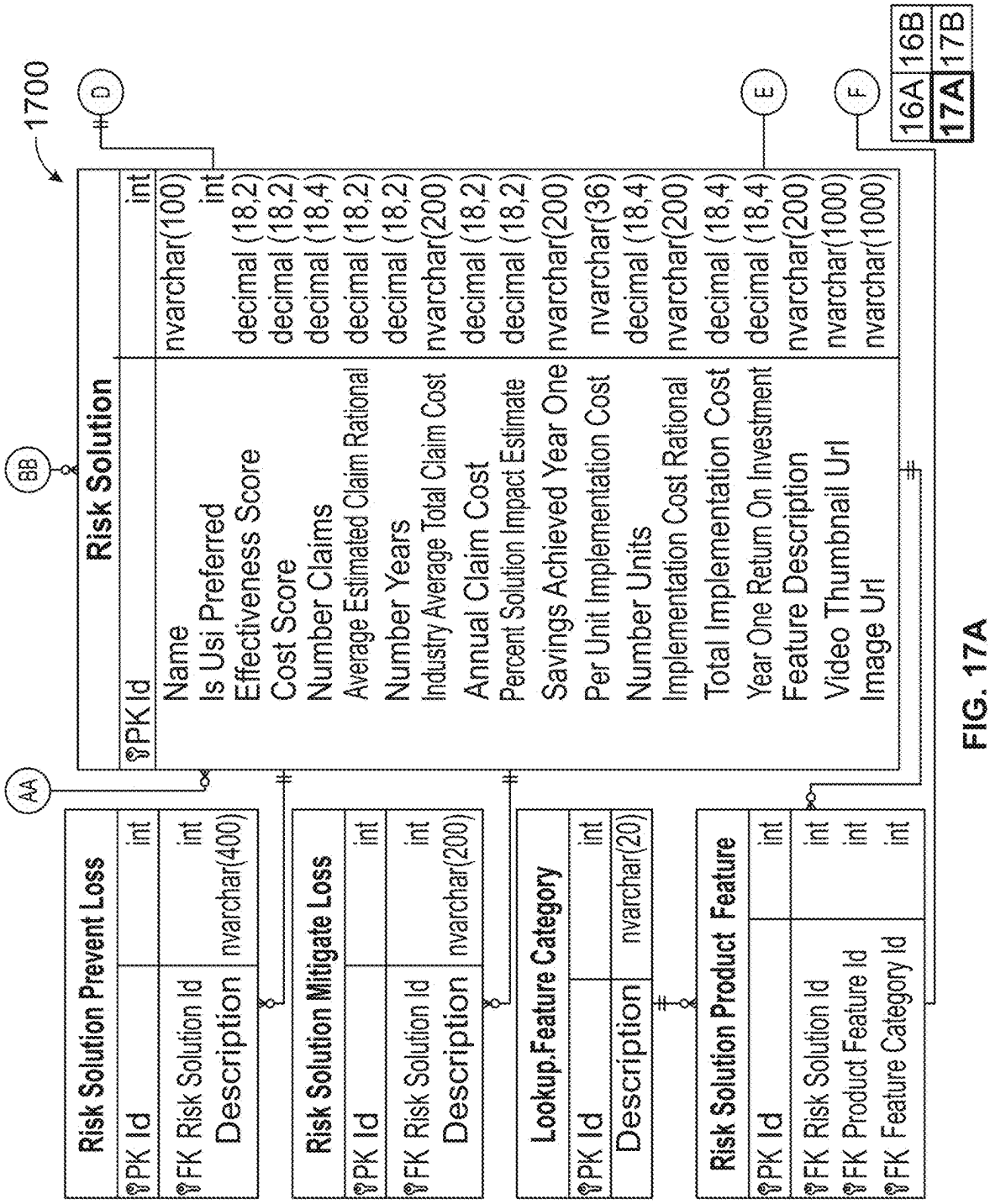

1700
Risk Solution
PK Id int
Name nvarchar(100)
Is Usi Preferred int
Effectiveness Score decimal (18,2)
Cost Score decimal (18,2)
Number Claims decimal (18,4)
Average Estimated Claim Rational decimal (18,2)
Number Years decimal (18,2)
Industry Average Total Claim Cost nvarchar(200)
Annual Claim Cost decimal (18,2)
Percent Solution Impact Estimate decimal (18,2)
Savings Achieved Year One nvarchar(200)
Per Unit Implementation Cost nvarchar(36)
Number Units decimal (18,4)
Implementation Cost Rational nvarchar(200)
Total Implementation Cost decimal (18,4)
Year One Return On Investment decimal (18,4)
Feature Description nvarchar(200)
Video Thumbnail Url nvarchar(1000)
Image Url nvarchar(1000)
Risk Solution Prevent Loss
PK Id int
FK Risk Solution Id int
Description nvarchar(400)
Risk Solution Mitigate Loss
PK Id int
FK Risk Solution Id int
Description nvarchar(200)
Lookup.Feature Category
PK Id int
Description nvarchar(20)
Risk Solution Product Feature
PK Id int
FK Risk Solution Id int
FK Product Feature Id int
FK Feature Category Id int
AA
BB
D
E
F
16A
16B
17A
17B

FIG. 17A

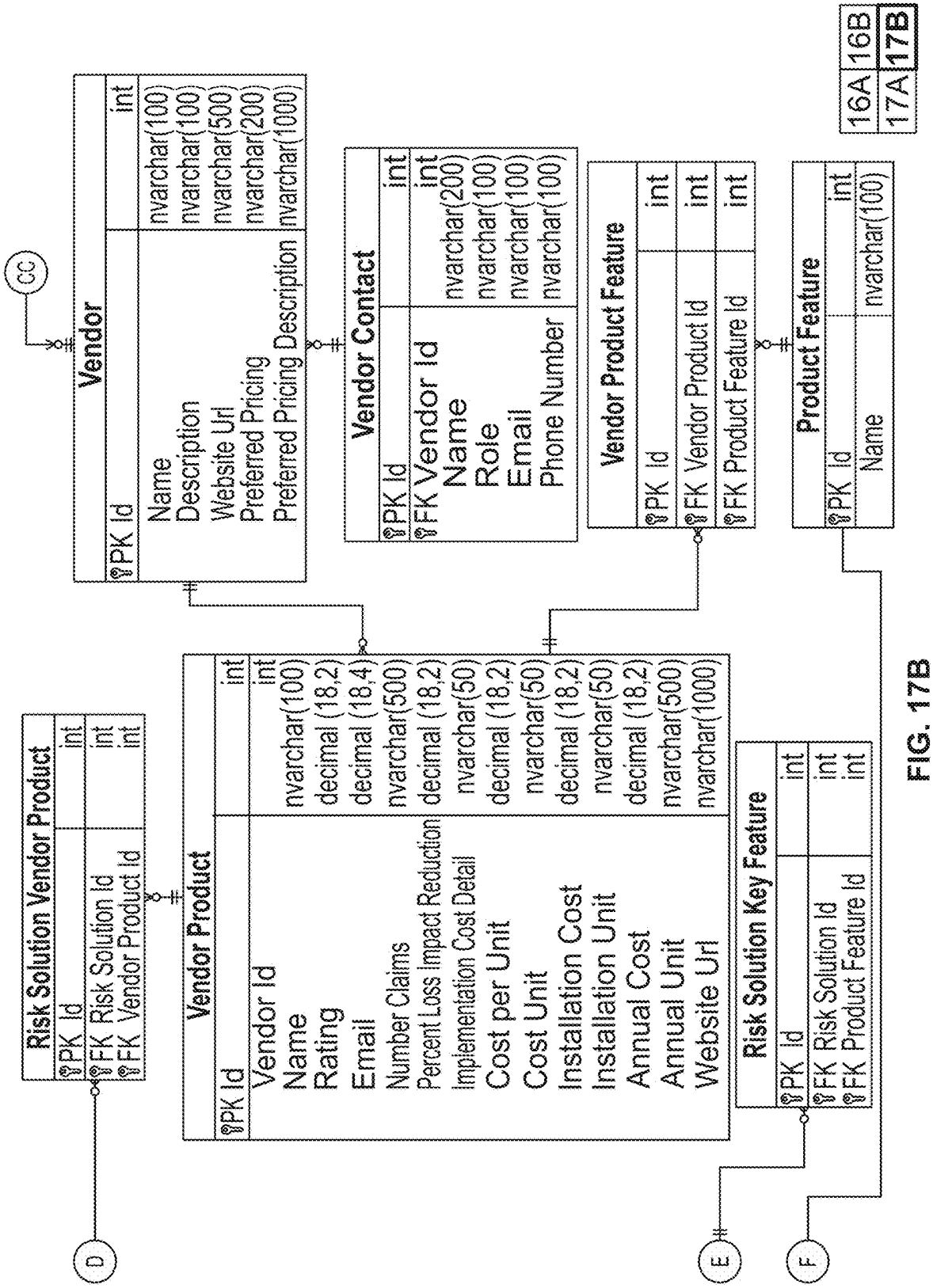
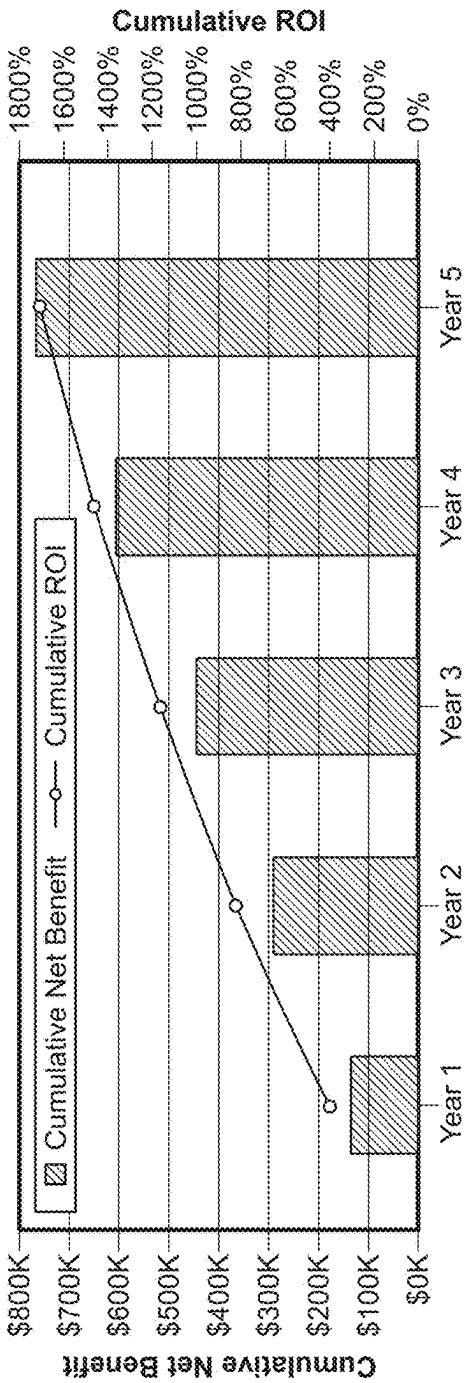

16A 16B
17A 17B
CC
Vendor
PK Id int
Name nvarchar(100)
Description nvarchar(100)
Website Url nvarchar(500)
Preferred Pricing nvarchar(200)
Preferred Pricing Description nvarchar(1000)
Vendor Contact
PK Id int
FK Vendor Id int
Name nvarchar(200)
Role nvarchar(100)
Email nvarchar(100)
Phone Number nvarchar(100)
Vendor Product Feature
PK Id int
FK Vendor Product Id int
FK Product Feature Id int
Product Feature
PK Id int
Name nvarchar(100)
Risk Solution Vendor Product
PK Id int
FK Risk Solution Id int
FK Vendor Product Id int
Vendor Product
PK Id int
Vendor Id int
Name nvarchar(100)
Rating decimal (18,2)
Email decimal (18,4)
Number Claims nvarchar(500)
Percent Loss Impact Reduction decimal (18,2)
Implementation Cost Detail nvarchar(50)
Cost per Unit decimal (18,2)
Cost Unit nvarchar(50)
Installation Cost decimal (18,2)
Installation Unit nvarchar(50)
Annual Cost decimal (18,2)
Annual Unit nvarchar(500)
Website Url nvarchar(1000)
Risk Solution Key Feature
PK Id int
FK Risk Solution Id int
FK Product Feature Id int
D
E
F

FIG. 17B

1800



Summary View Detailed View

Category		Y1	Y2	Y3	Y4	Y5
Investment	Implementation Costs	$32,000	-	-	-	-
	Ongoing Costs	-	$3,000	$3,000	$3,000	$3,000
	Cumulative Costs	$32,000	$35,000	$38,000	$41,000	$44,000
Return	Premium Savings	-	-	-	-	-
	Out Of Pocket Claims Avoidance	$180,000	$180,000	$180,000	$180,000	$180,000
	Indirect Costs Avoided	-	-	-	-	-
	Cumulative Savings	$180,000	$360,000	$540,000	$720,000	$900,000
	Cumulative Net Benefit	$148,000	$325,000	$502,000	$679,000	$856,000
	Cumulative ROI	460%	929%	1,321%	1,656%	1,945%
	IRR					649%

FIG. 18

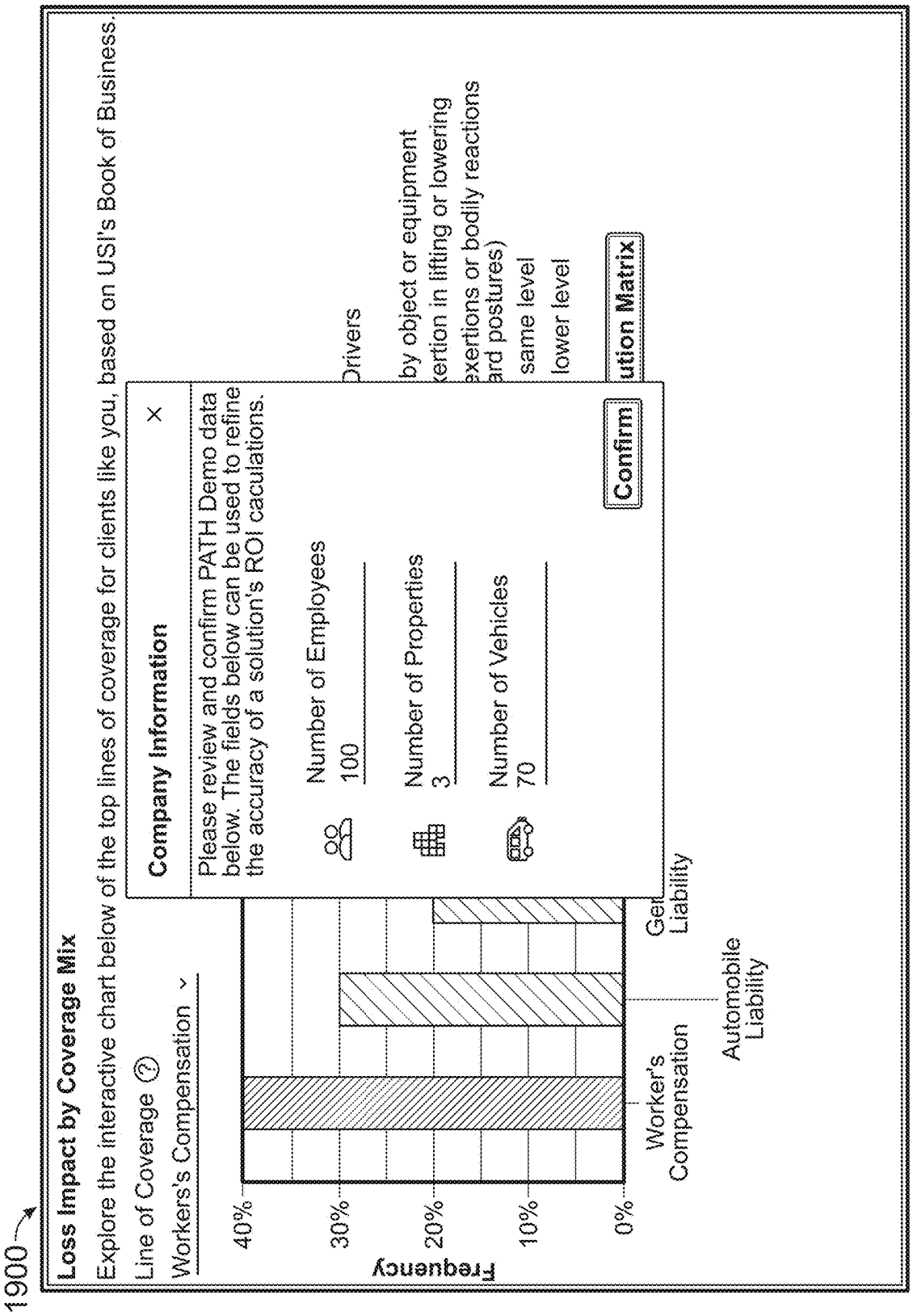

1900
Loss Impact by Coverage Mix
Explore the interactive chart below of the top lines of coverage for clients like you, based on USI's Book of Business.
Line of Coverage
Workers's Compensation
Company Information
Please review and confirm PATH Demo data below. The fields below can be used to refine the accuracy of a solution's ROI caculations.
Number of Employees
100
Number of Properties
3
Number of Vehicles
70
Confirm
Drivers
by object or equipment
kertion in lifting or lowering
exertions or bodily reactions
ard postures)
same level
lower level
ution Matrix
Frequency
40%
30%
20%
10%
0%
Worker's Compensation
Automobile Liability
Gen Liability

FIG. 19

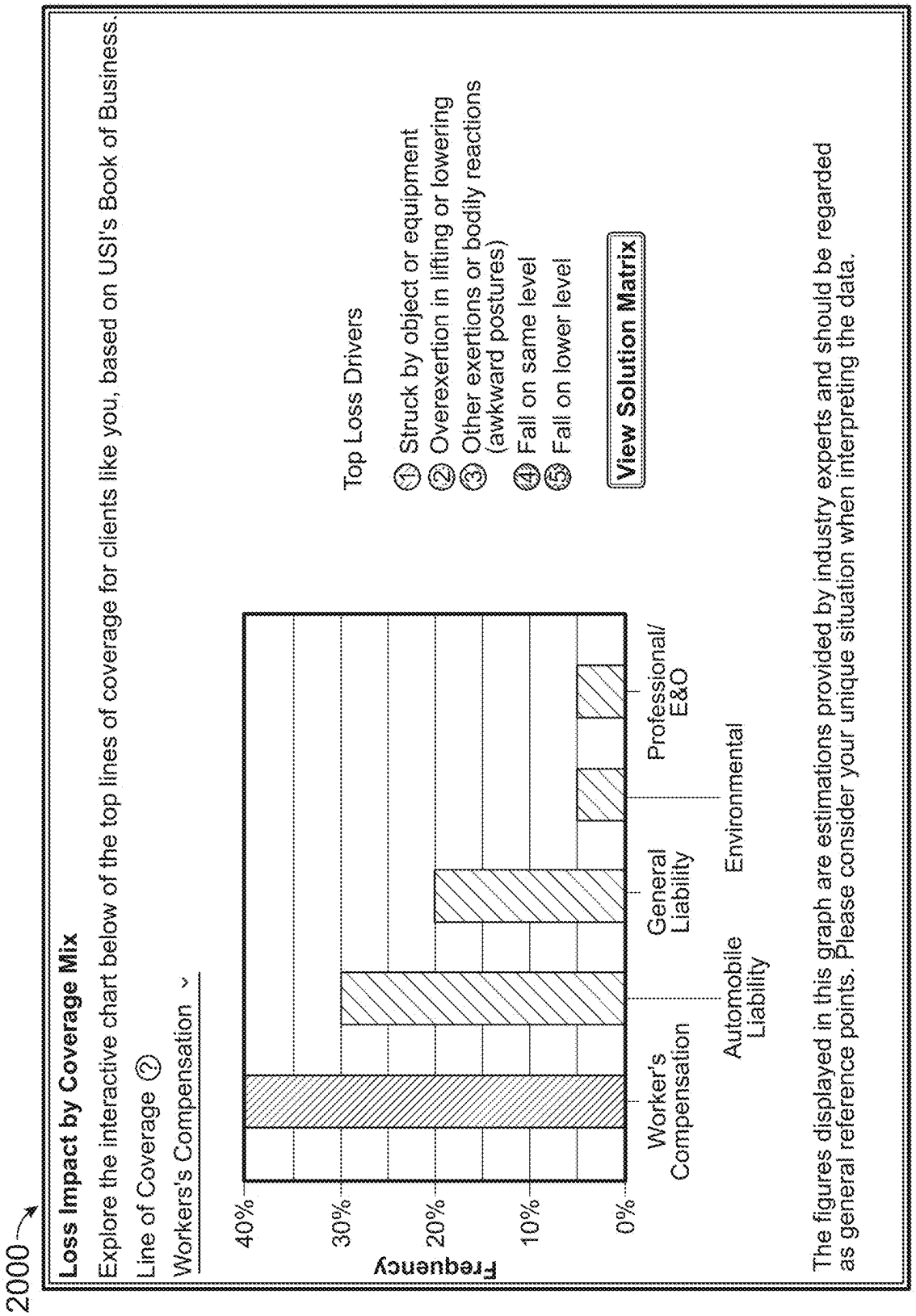

2000
Loss Impact by Coverage Mix
Explore the interactive chart below of the top lines of coverage for clients like you, based on USI's Book of Business.
Line of Coverage
Workers's Compensation
40%
30%
20%
10%
0%
Frequency
Worker's Compensation
Automobile Liability
General Liability
Environmental
Professional/ E&O
Top Loss Drivers
1 Struck by object or equipment
2 Overexertion in lifting or lowering
3 Other exertions or bodily reactions (awkward postures)
4 Fall on same level
5 Fall on lower level
View Solution Matrix
The figures displayed in this graph are estimations provided by industry experts and should be regarded as general reference points. Please consider your unique situation when interpreting the data.

FIG. 20

2100

PATH Summary Report

Below are the solutions and vendors you have added to your Summary, along with relevant information to support your review.
As a next step:

- Identify which solutions and vendors you wish to include for further consideration as part of a comprehensive risk control analysis and plan.
- Then, export to Excel for in-depth analysis and strategic planning.

Export Summary

Solutions (5) Vendors (4)

Industry
Construction

Subvertical
Contractors, Concrete Work

Line of Coverage
Workers' Compensation (2)

Solution Name	Loss Drivers	Annual Losses	Impact %	Year 1 Savings	Cost per Unit	# of Units	USI Discount	Actions
Fall Protection Program	(4) Fall on same level (5) Fall on lower level	$102,000	80%	$81,600	$200	60	0%	
Wearable Technology	(1) Struck by object or equipmen... (2) Overexertion in lifting or l... (3) Other exertions or bodily re...	$250,000	60%	$150,000	$400	154	0%	

FIG. 21

SYSTEM AND METHOD FOR COMPUTER BASED RISK CONTROL ANALYSIS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/561,148 filed on Mar. 4, 2024, titled "PATH Process," the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] In the insurance industry, effective claims handling and risk management may be critical to maintaining financial stability, ensuring customer trust, and complying with regulatory requirements. Claims processing may be a core function that directly impacts an insurer's profitability and reputation, as timely and fair settlements may enhance customer satisfaction while preventing fraudulent claims. Risk management may play an equally vital role by enabling insurers to assess, mitigate, and price risks accurately, helping them maintain a sustainable business model. With the increasing complexity of insurance products and regulatory frameworks, insurers may seek to streamline claims processing and improve risk assessment. Moreover, proactive risk management strategies, may allow insurers to anticipate potential losses and reduce unnecessary payouts.

[0003] By integrating robust claims handling processes with advanced risk management techniques, insurance companies may enhance operational efficiency, minimize financial losses, and provide policyholders with more reliable protection.

SUMMARY OF DISCLOSURE

[0004] In one example implementation, a computer-implemented method executed on a computing device may include but is not limited to, receiving client attribute data at a network server, identifying one or more relevant coverage lines based on the received client attribute data at the network server, and generating a solution matrix for each of the one or more relevant coverage lines at the network server, wherein the generated solution matrix is configured to be displayed on a graphical user interface (GUI).

[0005] One or more of the following example features may be included. Each solution matrix may include a set of possible solutions and each solution from the set of possible solutions may be represented as a point in the solution matrix. Each point in the solution matrix includes one or more loss drivers. In response to selecting a solution from the solution matrix, presenting a solution page for the selected solution on the GUI. The solution page may include more detailed information about the selected solution and one or more of a projection tool, a return on investment (ROI) analysis, an interactive calculator, and a vendor comparison. Identifying one or more vendors related to the selected solution at the network server, generating a vendor score for each vendor based on a set of viability factors at the network server, and generating a comparison table for the one or more vendors based on the generated vendor scores at the network server, where the generated comparison table may configured to be displayed on the GUI. The set of viability factors may include at least one of relevance of vendor products to the selected solution, history of performance, overall price, availability of discounts. In response to selecting a vendor from the comparison table, presenting a vendor page on the GUI. The vendor page includes one or more of an overview of vendor products and services, available discounts, and contact information.

[0006] In another example implementation, a computing system includes at least one processor and at least one memory architecture coupled with the at least one processor, where the at least one processor is configured to receive client attribute data at a network server, to identify one or more relevant coverage lines based on the received client attribute data at the network server, and to generate a solution matrix for each of the one or more relevant coverage lines at the network server, where the generated solution matrix may be configured to be displayed on a graphical user interface (GUI).

[0007] One or more of the following example features may be included. Each solution matrix may include a set of possible solutions, and each solution from the set of possible solutions is represented as a point in the solution matrix. Each point in the solution matrix includes one or more loss drivers. In response to selecting a solution from the solution matrix, presenting a solution page for the selected solution on the GUI. The solution page may include more detailed information about the selected solution and one or more of a projection tool, a return on investment (ROI) analysis, an interactive calculator, and a vendor comparison. The at least one processor may be further configured to identify one or more vendors related to the selected solution at the network server, generate a vendor score for each vendor based on a set of viability factors at the network server, and generate a comparison table for the one or more vendors based on the generated vendor scores at the network server, where the generated comparison table may be configured to be displayed on the GUI. The set of viability factors may include at least one of relevance of vendor products to the selected solution, history of performance, overall price, availability of discounts. In response to selecting a vendor from the comparison table, presenting a vendor page to be displayed on the GUI. The vendor page may include one or more of an overview of vendor products and services, available discounts, and contact information.

[0008] In another example implementation, a computer program product resides on a computer readable medium that has a plurality of instructions stored on it. When executed by a processor, the instructions cause the processor to perform operations that may include but are not limited to, receiving client attribute data at a network server, identifying one or more relevant coverage lines based on the received client attribute data at the network server, and generating a solution matrix for each of the one or more relevant coverage lines at the network server, where the generated solution matrix may be configured to be displayed on a graphical user interface (GUI).

[0009] One or more of the following example features may be included. Each solution matrix may include a set of possible solutions and each solution from the set of possible solutions may be represented as a point in the solution matrix.

[0010] The details of one or more example implementations are set forth in the accompanying drawings and the description below. Other possible example features and/or possible example advantages will become apparent from the description, the drawings, and the claims. Some implementations may not have those possible example features and/or

possible example advantages, and such possible example features and/or possible example advantages may not necessarily be required of some implementations.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is an example diagrammatic view of a risk control and analysis process coupled to a distributed computing network according to one or more example implementations of the disclosure;

[0012] FIG. **2** is an example flowchart of the risk control and analysis process of FIG. **1** according to one or more example implementations of the disclosure;

[0013] FIG. **3** is an example of a dashboard page according to one or more example implementations of the disclosure;

[0014] FIG. **4** is an example of a solution matrix according to one or more example implementations of the disclosure;

[0015] FIG. **5** is an example of a solution page according to one or more example implementations of the disclosure;

[0016] FIG. **6** is an example of an ROI calculator according to one or more example implementations of the disclosure;

[0017] FIG. **7** is an example of a projection tool according to one or more example implementations of the disclosure;

[0018] FIG. **8** is an example of a vendor comparison according to one or more example implementations of the disclosure;

[0019] FIG. **9** is an example of a vendor page according to one or more example implementations of the disclosure; and

[0020] FIGS. **10-21** are example diagrammatic views of the risk control and analysis process of FIG. **1** according to one or more example implementations of the disclosure.

[0021] Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

System Overview

[0022] Referring to FIG. **1**, there is shown risk control and analysis process **10** that may reside on and may be executed by server computer **12**, which may be connected to network **14** (e.g., the internet or a local area network). Examples of server computer **12** may include, but are not limited to: a personal computer, a server computer, a series of server computers, a mini computer, and a mainframe computer. Server computer **12** may be a web server (or a series of servers) running a network operating system, examples of which may include but are not limited to: Microsoft Windows XP Server™; Novell Netware™; or Redhat Linuxt™, for example. Additionally and/or alternatively, the routing topology process may reside on a client electronic device, such as a personal computer, notebook computer, personal digital assistant, or the like.

[0023] The instruction sets and subroutines of the risk control and analysis process **10**, which may be stored on storage device **16** coupled to server computer **12**, may be executed by one or more processors (not shown) and one or more memory architectures (not shown) incorporated into server computer **12**. Storage device **16** may include but is not limited to: a hard disk drive; a tape drive; an optical drive; a RAID array; a random access memory (RAM); and a read-only memory (ROM).

[0024] Server computer **12** may execute a web server application, examples of which may include but are not limited to: Microsoft IIS™, Novell Webserver™, or Apache Webserver™, that allows for HTTP (i.e., HyperText Transfer Protocol) access to server computer **12** via network **14**. Network **14** may be connected to one or more secondary networks (e.g., network **18**), examples of which may include but are not limited to: a local area network; a wide area network; or an intranet, for example.

[0025] Server computer **12** may execute one or more server applications (e.g., server application **20**), examples of which may include but are not limited to, e.g., Microsoft Exchange™ Server, etc. Server application **20** may interact with one or more client applications (e.g., client applications **22**, **24**, **26**, **28**) in order to execute risk control and analysis process **10**. Examples of client applications **22**, **24**, **26**, **28** may include, but are not limited to, EDAs or design verification tools such as those available from the assignee of the present disclosure. These applications may also be executed by server computer **12**. In some embodiments, risk control and analysis process **10** may be a stand-alone application that interfaces with server application **20** or may be applets/applications that may be executed within server application **20**.

[0026] The instruction sets and subroutines of server application **20**, which may be stored on storage device **16** coupled to server computer **12**, may be executed by one or more processors (not shown) and one or more memory architectures (not shown) incorporated into server computer **12**.

[0027] As mentioned above, in addition, or as an alternative to being server-based applications residing on server computer **12**, risk control and analysis process **10** may be a client-side application residing on one or more client electronic devices **38**, **40**, **42**, **44** (e.g., stored on storage devices **30**, **32**, **34**, **36**, respectively). As such, risk control and analysis process **10** may be a stand-alone application that interfaces with a client application (e.g., client applications **22**, **24**, **26**, **28**), or may be applets/applications that may be executed within a client application. As such, risk control and analysis process **10** may be a client-side process, server-side process, or hybrid client-side/server-side process, which may be executed, in whole or in part, by server computer **12**, or one or more of client electronic devices **38**, **40**, **42**, **44**.

[0028] The instruction sets and subroutines of client applications **22**, **24**, **26**, **28**, which may be stored on storage devices **30**, **32**, **34**, **36** (respectively) coupled to client electronic devices **38**, **40**, **42**, **44** (respectively), may be executed by one or more processors (not shown) and one or more memory architectures (not shown) incorporated into client electronic devices **38**, **40**, **42**, **44** (respectively). Storage devices **30**, **32**, **34**, **36** may include but are not limited to: hard disk drives; tape drives; optical drives; RAID arrays; random access memories (RAM); read-only memories (ROM), compact flash (CF) storage devices, secure digital (SD) storage devices, and memory stick storage devices. Examples of client electronic devices **38**, **40**, **42**, **44** may include, but are not limited to, personal computer **38**, laptop computer **40**, personal digital assistant **42**, notebook computer **44**, a data-enabled, cellular telephone (not shown), and a dedicated network device (not shown), for example. Using client applications **22**, **24**, **26**, **28**, users **46**, **48**, **50**, **52** may utilize the EDA to create an electronic design.

[0029] Users **46**, **48**, **50**, **52** may access server application **20** directly through the device on which the client application (e.g., client applications **22**, **24**, **26**, **28**) is executed, namely client electronic devices **38**, **40**, **42**, **44**, for example. Users **46**, **48**, **50**, **52** may access server application **20** directly through network **14** or through secondary network **18**. Further, server computer **12** (e.g., the computer that executes server application **20**) may be connected to network **14** through secondary network **18**, as illustrated with phantom link line **54**.

[0030] In some embodiments, risk control and analysis process **10** may be a cloud-based process as any or all of the operations described herein may occur, in whole, or in part, in the cloud or as part of a cloud-based system. The various client electronic devices may be directly or indirectly coupled to network **14** (or network **18**). For example, personal computer **38** is shown directly coupled to network **14** via a hardwired network connection. Further, notebook computer **44** is shown directly coupled to network **18** via a hardwired network connection. Laptop computer **40** is shown wirelessly coupled to network **14** via wireless communication channel **56** established between laptop computer **40** and wireless access point (i.e., WAP) **58**, which is shown directly coupled to network **14**. WAP **58** may be, for example, an IEEE 802.11a, 802.11b, 802.11g, Wi-Fi, and/or Bluetooth device that is capable of establishing wireless communication channel **56** between laptop computer **40** and WAP **58**. Personal digital assistant **42** is shown wirelessly coupled to network **14** via wireless communication channel **60** established between personal digital assistant **42** and cellular network/bridge **62**, which is shown directly coupled to network **14**.

[0031] As is known in the art, all of the IEEE 802.11x specifications may use Ethernet protocol and carrier sense multiple access with collision avoidance (CSMA/CA) for path sharing. The various 802.11x specifications may use phase-shift keying (PSK) modulation or complementary code keying (CCK) modulation, for example. As is known in the art, Bluetooth is a telecommunications industry specification that allows e.g., mobile phones, computers, and personal digital assistants to be interconnected using a short-range wireless connection.

[0032] Client electronic devices **38**, **40**, **42**, **44** may each execute an operating system, examples of which may include but are not limited to Microsoft Windows™, Microsoft Windows CE™, Redhat Linux™, Apple IOS, ANDROID, or a custom operating system.

The Risk Control and Analysis Process

[0033] Referring also to FIGS. **2-9** and in some implementations, risk control and analysis process **10** may receive (**202**) client attribute data at a network server, identify (**204**) one or more relevant coverage lines based on the received client attribute data at the network server, and generate (**206**) a solution matrix for each of the one or more relevant coverage lines at the network server, where the generated solution matrix may be configured to be displayed on a graphical user interface (GUI). In response to selecting a solution from the solution matrix, risk control and analysis process **10** may further present (**208**) a solution page for the selected solution on the GUI, identify (**210**) one or more vendors related to the selected solution at the network server, and generate (**212**) a vendor score for each vendor based on a set of viability factors at the network server. Risk control and analysis process **10** may also generate (**214**) a comparison table for the one or more vendors based on the generated vendor scores at the network server, where the generated comparison table may be configured to be displayed on the GUI. In response to selecting a vendor from the comparison table, risk control and analysis process **10** may present (**216**) a vendor page on the GUI.

[0034] In some implementations, risk control and analysis process **10** may receive (**202**) client attribute data and identify (**204**) one or more relevant coverage lines based on the received client attribute data. Client attribute data may include factors like the products and services offered by the client (i.e. industry), the market share, the number of branch offices, the location of each office, the number of employees, etc. A coverage line may be a specific type or category of insurance coverage offered within an insurance policy or program. Coverage lines may be used to distinguish between different areas of risk or protection and may align with various aspects of a client's needs.

[0035] Consider for example, dashboard page **300** shown in FIG. **3**. Dashboard page **300** may include a graphical depiction of one or more of the most relevant coverage lines, e.g. bar chart **302**, and a corresponding list of loss drivers (e.g. listing **304**). Loss drivers may refer to primary factors or underlying causes that contribute significantly to claims, financial losses, or risks within an organization. Identifying and understanding loss drivers may be critical for developing targeted strategies to mitigate risks and reduce the frequency or severity of claims. Data about the loss drivers may be sourced from aggregated claims data and industry experts. In some implementations, risk control and analysis process **10** may use a database of aggregated and anonymized client data to inform actual top lines of coverage and loss drivers within the data set. Loss drivers may still be augmented with additional industry data as needed. Futher, users may choose to customize their experience by uploading their loss runs (claims data) or by selecting their actual top loss drivers. Dashboard page **300** may also include one or more dropdown menus for additional filtering (e.g. industry **306** or subverticals **308**). End-users may also use dashboard page **300** to explore the top causes of loss for other coverage lines by toggling through one or more tabs provided, for example ribbon **310** may include tabs for worker's compensation, auto liability, general liability, etc.

[0036] In some implementations, risk control and analysis process **10** may generate (**206**) a solution matrix for each of the one or more relevant coverage lines shown in bar chart **302**. Each solution matrix may include a set of possible solutions and each solution from the set of possible solutions may be represented as a point in the solution matrix. Further, each point/solution in the solution matrix may be associated with one or more loss drivers. More generally, solutions may be identified by subject matter experts (SMEs) and through research of emerging technologies. Solutions may be associated with the loss drivers through review by SMEs, and solution matrix coordinates may be the result of input by SMEs. In some implementations, risk control and analysis process **10** may use a database of aggregated and anonymized client data that may include the captured results from solution/vendor results, which may further inform the solution matrix.

[0037] Consider, for example, dashboard page **400** shown in FIG. **4**. The solution set for each coverage line may be plotted in a cost-effectiveness matrix, like solution matrix

402 plotted for the worker's compensation coverage line **404**. End-users may examine solutions included in the set of possible solutions by hovering a cursor over solution matrix **402** or by examining solutions table **406**, for example, the fall protection program solution **408** and a corresponding list of loss drivers. End-users may further filter what solutions are shown by selecting a loss driver from among the one or more loss drivers presented in loss driver ribbon **410**. End-users may also toggle between different coverage lines by selecting from among the one or more coverage line tabs provided in coverage line dropdown menu.

[0038] In some implementations, in response to selecting a solution from the solution matrix, risk control and analysis process **10** may further present (**208**) a solution page, providing an overview of the potential solution. Consider, for example, solution page **500** for a wearable technology solution. Solution page **500** may include one or more images, videos, or other media (e.g. image **502**) that may provide end-users with a better understanding of how the wearable technology solution works, what it may look like, or how it may be used. Solution page **500** may also include a brief synopsis of features (e.g. one or more features **504**) or a brief synopsis of industry-specific loss drivers (e.g. one or more loss drivers **506**). Further, solution page **500** may also provide a brief synopsis of possible benefits provided by the wearable technology solution.

[0039] In some implementations, risk control and analysis process **10** may further identify (**210**) one or more vendors related to the selected solution, generate (**212**) a vendor score for each vendor based on a set of viability factors, and generate (**214**) a comparison table for the one or more vendors based on the generated vendor scores. Vendors in this context may refer to any supplier of the products or services used in the selected solution. For example, in the previously discussed wearable technology solution, a vendor could be any provider of smart watches, smart glasses, etc. Vendor scores may represent a ranking system that end-users may use as a basis for comparison between vendors. Vendor scores may be based on a set of viability factors including, but not limited to: (i) relevance of vendor products to the selected solution, (ii) history of performance, (iii) overall price, and (iv) availability of discounts. Vendor scores may also be achieved via a weighted average of the identified feature set of a solution and a vendor's ability to provide each feature. In some implementations, risk control and analysis process **10** may use a database of aggregated and anonymized client data that may include the captured results from solution/vendor results, which may allow for additional scoring based on actual results generated for end-users.

[0040] In some implementations, solution pages may include one or more of an interactive calculator for estimating first-year return on investment (ROI), an ROI projection tool, and a vendor comparison. Consider, for example, solution page **600**, shown in FIG. **6**. Solution page **600** may allow end-users to perform ROI analyses with an interactive calculator based on the client company's history of loss and apparent solution needs. On the returns side, the interactive calculator may consider factors like: (i) estimated losses, (ii) industry average cost per loss, (iii) solution impact, and (iv) indirect cost savings. On the investment side, the interactive calculator may consider factors like: (i) cost per unit, (ii) number of units, and (iii) percentage discount if any. The interactive calculator may compute a first year loss sensitive ROI based on implementation costs for that solution and claims data.

[0041] Consider for example solution page **700** shown in FIG. **7**, which may provide end-users with a five-year projection tool (e.g. ROI projection **702**) that considers the type of funding and carrier credits to graph the cumulative net benefit and ROI of implementing the solution. In some implementations, the 5-year ROI projection may model the dollar savings, and consider the funding type (guaranteed cost or deductible) and carrier credits. These models may be created with SMEs to show the potential impact of reduced losses on future premiums paid. The ROI model may be further refined using a database of aggregated/anonymized client data, actual vendor results for those clients, and a claims benchmarking tool. Such data may allow to have pre-negotiated credits with specific carriers. Risk control and analysis process **10** may use a database of aggregated and anonymized client data that may include the captured results from solution/vendor results, which may allow risk control and analysis process **10** to generate additional scoring based on actual results generated for end-users. End-users may also view a summary and detailed breakdown of the projection (see FIG. **18**).

[0042] Consider for example solution page **800** shown in FIG. **8**, which may provide end-users with a vendor comparison table (e.g. vendor table **802**) that may show several vendors side-by-side, along with which features they provide and which features they do not provide. Vendor table **802** may also provide the generated vendor scores and indicate which vendors if any provide in-network discounted prices for added convenience. Where the amount discounted may vary based on the individual vendor involved. Solution page **800** may also allow end-users to select one or more vendors for comparison from a pool of available vendors (vendor list **804**).

[0043] In response to selecting a vendor from the comparison table, in some implementations risk control and analysis process **10** may further present (**216**) a vendor page. End-users may view a vendor page by selecting a vendor from vendor table **802**. Consider, for example, vendor page **900** shown in FIG. **9**, which may provide an overview (e.g. overview **902**) of the selected vendor and the products and/or services that they offer. The solutions offered by the vendor may be detailed in a carousel menu (e.g. solutions menu **904**), where end-users may browse pricing and feature information. Vendors who elect to offer discounted pricing may also provide information (e.g. discount steps **906**) on how to redeem the discount on vendor page **900**. Similarly, if a vendor elects to display point-of-contact information for an agent, the agent's contact details (e.g. contact info **908**) may also be provided on vendor page **900**.

[0044] Referring now to FIG. **10**, webpage **1000** is provided with details of a coverage line consistent with embodiments of the present disclosure. Webpage **1000** may provide a deeper understanding of an "Auto-Liability" coverage line for a company called "ABC Widget Design". Moreover, webpage **1000** may provide a breakdown of loss impact by coverage mix, which gives a graph of loss impact information and a corresponding list of loss drivers for each coverage line. End-users may examine similar graphs and loss driver lists for each coverage line.

[0045] Referring now to FIG. **11**, webpage **1100** is provided showing an overview of a solution page for an "AI

Dashboard Camera". The overview may provide detailed information about the features of the solution, and end-users may also view details on the associated benefits and loss drivers for the solution. Additionally, webpage **1100** may include a one year return on investment (ROI) calculator, pre-populated with data relevant to the solution being considered.

[0046] Referring now to FIG. **12**, webpage **1200** is provided showing an overview of a solution page for an "AI Dashboard Camera". Webpage **1200** may continue with the example of the "Auto-Liability" coverage line for "ABC Widget Design". In particular, webpage **1200** may show a solution matrix, where the "AI Dashboard Camera" solution previously discussed in FIG. **11** is selected. The bottom of webpage **1200** may show a comparison of solutions for the auto-liability coverage line, of which the AI dashboard camera was ranked highest with an estimated 80% solution impact, which may be a measure of the effectiveness of the solution.

[0047] Referring now to FIGS. **13**A-C, a single webpage **1300** is provided, but due to practical restrictions on image size and quality has been broken down and spread across three pages. Webpage **1300** may provide details for a "Wearable Technology" solution. For example, FIG. **13**A may show a listing of industry-specific loss drivers and a one year ROI calculator, pre-populated with data relevant to the solution being considered. FIG. **13**A may also show where end-users are able to view details on the features or benefits of the wearable technology solution. FIG. **13**B may provide a 5-year ROI projection of how adopting the wearable technology solution might play out, and may also show where end-users may access either a summary view or a detailed view of the solution impact for each year over the 5 year projection. FIG. **13**C may provide a comparison of vendors who may provide the solution being considered, in this case the wearable technology solution. The vendor comparison my place each vendor side by side and give end-users the opportunity to clearly see what features each of the solution each vendor is able to provide and which features if any they do not provide.

[0048] Referring now to FIG. **14**, a webpage **1400** is provided showing an overview of a selected vendor and the products and/or services that they offer. In this case the vendor may be a company called "Monnit" and webpage **1400** may provide details for each solution offered by the vendor, such that end-users may browse pricing and feature information. Vendors who elect to offer discounted pricing may also provide information on how to redeem the discount. Vendors may also elect to display point-of-contact information for an agent, the agent's contact details may also be provided.

[0049] Referring now to FIGS. **15-17**, a risk control and analysis process conceptual data model **1500**, a risk control and analysis process schema **1600**, and a risk control and analysis lookup schema **1700** are provided. Risk control and analysis process schema **1600** may be a structured framework that organizes and stores comprehensive data related to risk assessment. This schema may capture key metrics such as risk factors, policyholder data, historical claims, and risk exposure to provide actionable insights for underwriting and claims management. On the other hand, risk control and analysis lookup schema **1700** may serve as a reference system that stores predefined risk parameters, industry benchmarks, and standard risk categories, allowing for efficient cross-referencing and comparison with data in the process schema. These two schemas may be closely related, as the lookup schema provides static, reference data that enhances the dynamic, transactional data within the process schema, enabling more accurate and informed decision-making throughout the risk control and analysis process. Together, process schema **1600** and lookup schema **1700** ensure that risk analysis may be both comprehensive and adaptable to evolving data.

[0050] Referring now to FIG. **18**, webpage **1800** is provided showing a 5-year ROI projection of how adopting another solution might play out, and may also show where end-users may access either a summary view or a detailed view of the solution impact for each year over the 5 year projection.

[0051] Referring now to FIG. **19**, webpage **1900** is provided showing where an end-user may be prompted to confirm firmographic data. This information, paired with backend industry, solution, and vendor data, may be used to provide automatic solution ROI estimates with little to no intervention from the user.

[0052] Referring now to FIG. **20**, webpage **2000** is provided showing a breakdown of loss impact by coverage mix, which gives a graph of loss impact information and a corresponding list of loss drivers for each coverage line. End-users may examine similar graphs and loss driver lists for each coverage line. Webpage **2000** may present end-users with the top coverage lines and respective top causes of loss for their industry, such that the user may explore different industries and subverticals. The end-user may also explore the top causes of loss for other coverage lines in the dropdown menu.

[0053] Referring now to FIG. **21**, example **2100** is provided showing a summary report for one or more solutions selected by an end-user, where the end-user may add solutions with customized ROIs and vendors to a summary page. Furthermore, the summary report may be exported to Excel file for more in-depth analysis and strategic planning.

[0054] In some implementations, risk control and analysis process **10** may provide an automated extraction of data from various document formats, including unstructured or semi-structured documents like PDFs and structured documents such as Excel and Comma Separated Values (CSV) files. Risk control and analysis process **10** may guide these documents through a unified pipeline, while efficiently extracting business-defined data and storing it in an enterprise data warehouse. This robust solution may enhance data management and utilization, making it highly effective for diverse applications. Document Insights may take as input, a file listing insurance claims (client attribute data) and return the data in a structured format for analysis.

[0055] In some implementations, the data extraction process may include three major stages: pre-processing, parsing, and validation. The pre-processing stage may involve the initial cleaning and preparation of documents to enhance data readability. The parsing stage may involve utilizing specialized algorithms to interpret and extract relevant data fields from each document type. The validation stage may involve ensuring the accuracy and completeness of extracted data through cross-referencing and error-checking mechanisms.

[0056] In some implementations, risk control and analysis process **10** may assemble all extracted and integrated data into a single javascript object notation (JSON) output. This

JSON file may serve as a comprehensive representation of the processed data, ready for further analysis or application. Once extracted, the data may be transformed into a structured format and integrated into an enterprise data warehouse. More specifically, the data may efficiently upload the transformed data into a cloud-based data warehouse platform, ensuring minimal latency and data integrity, then the extracted data may be converted into a standardized format suitable for use in downstream applications. Risk control and analysis process **10** may be designed to be easily implemented and scaled, allowing for adaptation to various document types.

[0057] In some implementations, risk control and analysis process **10** may provide numerous advantages over existing approaches. Some of these may include but are not limited to, the concepts identified below. Aggregation of industry benchmarks, client loss data, and intellectual capital that provides the client with a self-service risk control platform. A curated list of solutions and relevant vendors tailored to the client's industry and loss drivers supported with ROI projections and preferred vendor pricing. 5-Year premium projections provide leverage proprietary assumptions for different program structures. Negotiated premium credits with select carriers. Comparison grid of top vendors and prioritized features for each solution.

[0058] In some implementations, by integrating industry benchmarks, clients may compare their risk profile against industry standards, identifying strengths and weaknesses in their risk management practices. Clients may also gain data-driven insights on how their risk exposure may compare to their peers, helping them make more informed decisions. Client loss data may enable users to identify trends and patterns in their own claims history, thereby helping them to proactively address recurring risks. With predictive analytics, risk control and analysis process **10** may offer customized recommendations to reduce future losses.

[0059] In some implementations, by leveraging intellectual capital (e.g., expert insights, proprietary risk models, best practices), risk control and analysis process **10** may provide automated guidance tailored to the client's specific industry and risk factors. Clients may access expert-level risk control recommendations without needing an in-person consultant, saving time and money. Moreover, a self-service model may allow clients to independently access risk analysis tools, reducing reliance on manual intervention.

[0060] In some implementations, real-time analytics and dashboards may provide instant feedback on risk exposure, empowering clients to make timely adjustments. Risk control and analysis process **10** may also be adapted to different industries, risk tolerances, and business sizes, making it scalable for a variety of clients. Further, clients may set parameters and tailor reports based on their unique risk environment. By minimizing manual risk assessments and consultant dependencies, clients may be able reduce operational costs. The automation of risk control and analysis process **10** may ensure consistent, accurate, and repeatable assessments. Additionally risk control and analysis process **10** may help clients stay compliant with industry regulations by maintaining up-to-date risk management practices. Centralized data storage may ensure easy access to risk documentation and audit trails for regulatory reporting.

[0061] In some embodiments, risk control and analysis process **10** may provide the ability to ingest loss run data from carriers for an even more personalized experience. The process may provide a vendor portal or access point for third-parties to make edits to their pages and product info directly. The process may export a "Risk Control Plan" report. The process may also provide Search, favoriting, and other helpful features to enhance user experience. The process may additionally provide analytics that measure the impact of risk control solutions on client's actual loss costs and premiums over time.

[0062] It will be apparent to those skilled in the art that various modifications and variations can be made to PATH process **10** and/or embodiments of the present disclosure without departing from the spirit or scope of the invention. Thus, it is intended that embodiments of the present disclosure cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

General

[0063] As will be appreciated by one skilled in the art, the present disclosure may be embodied as a method, a system, or a computer program product. Accordingly, the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a "circuit," "module" or "system." Furthermore, the present disclosure may take the form of a computer program product on a computer-usable storage medium having computer-usable program code embodied in the medium.

[0064] Any suitable computer usable or computer readable medium may be utilized. The computer-usable or computer-readable medium may be, for example but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, device, or propagation medium. More specific examples (a non-exhaustive list) of the computer-readable medium may include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), an optical storage device, a transmission media such as those supporting the Internet or an intranet, or a magnetic storage device. The computer-usable or computer-readable medium may also be paper or another suitable medium upon which the program is printed, as the program can be electronically captured, via, for instance, optical scanning of the paper or other medium, then compiled, interpreted, or otherwise processed in a suitable manner, if necessary, and then stored in a computer memory. In the context of this document, a computer-usable or computer-readable medium may be any medium that can contain, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device. The computer-usable medium may include a propagated data signal with the computer-usable program code embodied therewith, either in baseband or as part of a carrier wave. The computer usable program code may be transmitted using any appropriate medium, including but not limited to the Internet, wireline, optical fiber cable, RF, etc.

[0065] Computer program code for carrying out operations of the present disclosure may be written in an object oriented programming language such as Java, Smalltalk, C++ or the like. However, the computer program code for carrying out operations of the present disclosure may also be written in conventional procedural programming languages, such as the "C" programming language or similar programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through a local area network/a wide area network/the Internet (e.g., network **14**).

[0066] The present disclosure is described with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to implementations of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, may be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer/special purpose computer/other programmable data processing apparatus, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0067] These computer program instructions may also be stored in a computer-readable memory that may direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture including instruction means which implement the function/act specified in the flowchart and/or block diagram block or blocks.

[0068] The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0069] The flowcharts and block diagrams in the figures may illustrate the architecture, functionality, and operation of possible implementations of systems, methods and computer program products according to various implementations of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, may be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0070] The terminology used herein is for the purpose of describing particular implementations only and is not intended to be limiting of the disclosure. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0071] The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the disclosure in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiment was chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various implementations with various modifications as are suited to the particular use contemplated.

[0072] A number of implementations have been described. Having thus described the disclosure of the present application in detail and by reference to implementations thereof, it will be apparent that modifications and variations are possible without departing from the scope of the disclosure defined in the appended claims.

What is claimed is:

1. A computer-implemented method executed on a computing device for risk control analysis, comprising:

receiving client attribute data at a network server;

identifying one or more relevant coverage lines based on the received client attribute data at the network server; and

generating a solution matrix for each of the one or more relevant coverage lines at the network server, wherein the generated solution matrix is configured to be displayed on a graphical user interface (GUI).

2. The computer-implemented method of claim **1**, wherein each solution matrix includes a set of possible solutions and each solution from the set of possible solutions is represented as a point in the solution matrix.

3. The computer-implemented method of claim **2**, wherein each point in the solution matrix includes one or more loss drivers.

4. The computer-implemented method of claim **2**, further including:

in response to selecting a solution from the solution matrix, presenting a solution page for the selected solution on the GUI.

5. The computer-implemented method of claim **4**, wherein the solution page includes more detailed information about the selected solution and one or more of a

projection tool, a return on investment (ROI) analysis, an interactive calculator, and a vendor comparison.

6. The computer-implemented method of claim **4**, further including:

identifying one or more vendors related to the selected solution at the network server;

generating a vendor score for each vendor based on a set of viability factors at the network server; and

generating a comparison table for the one or more vendors based on the generated vendor scores at the network server, wherein the generated comparison table is configured to be displayed on the GUI.

7. The computer-implemented method of claim **6**, wherein the set of viability factors include at least one of relevance of vendor products to the selected solution, history of performance, overall price, availability of discounts.

8. The computer-implemented method of claim **6**, further including:

in response to selecting a vendor from the comparison table, presenting a vendor page on the GUI.

9. The computer-implemented method of claim **8**, wherein the vendor page includes one or more of an overview of vendor products and services, available discounts, and contact information.

10. A computing system comprising:

a memory; and

a processor configured to receive client attribute data at a network server, to identify one or more relevant coverage lines based on the received client attribute data at the network server, and to generate a solution matrix for each of the one or more relevant coverage lines at the network server, wherein the generated solution matrix is configured to be displayed on a graphical user interface (GUI).

11. The computing system of claim **10**, wherein each solution matrix includes a set of possible solutions, and each solution from the set of possible solutions is represented as a point in the solution matrix.

12. The computing system of claim **11**, wherein each point in the solution matrix includes one or more loss drivers.

13. The computing system of claim **11**, further including:

in response to selecting a solution from the solution matrix, presenting a solution page for the selected solution on the GUI.

14. The computing system of claim **13**, wherein the solution page includes more detailed information about the selected solution and one or more of a projection tool, a return on investment (ROI) analysis, an interactive calculator, and a vendor comparison.

15. The computing system of claim **13**, wherein the processor is further configured to:

identify one or more vendors related to the selected solution at the network server;

generate a vendor score for each vendor based on a set of viability factors at the network server; and

generate a comparison table for the one or more vendors based on the generated vendor scores at the network server, wherein the generated comparison table is configured to be displayed on the GUI.

16. The computing system of claim **15**, wherein the set of viability factors include at least one of relevance of vendor products to the selected solution, history of performance, overall price, availability of discounts.

17. The computing system of claim **15**, further including:

in response to selecting a vendor from the comparison table, presenting a vendor page to be displayed on the GUI.

18. The computing system of claim **17**, wherein the vendor page includes one or more of an overview of vendor products and services, available discounts, and contact information.

19. A computer program product residing on a non-transitory computer readable medium having a plurality of instructions stored thereon which, when executed by a processor, cause the processor to perform operations comprising:

receiving client attribute data at a network server;

identifying one or more relevant coverage lines based on the received client attribute data at the network server; and

generating a solution matrix for each of the one or more relevant coverage lines at the network server, wherein the generated solution matrix is configured to be displayed on a graphical user interface (GUI).

20. The computer program product of claim **19**, wherein each solution matrix includes a set of possible solutions, and each solution from the set of possible solutions is represented as a point in the solution matrix.

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